

penev_A — formula report

674 inline formulas (section omitted; 1 did not render), 116 display equations, 2 tables, 25 unrecovered image regions.

Display equations

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
penev_A_ E00041 (2.44)	008	■ 0.244 ● W +0	$\begin{aligned} \Phi &= \sum_r \sigma_r \mathbf{s}_r(\psi_r,) \\ \Phi^* \Phi &= \mathbf{R} = \sum_r \sigma_r^2(\psi_r,) \\ \mathrm{tr} \mathbf{R} &= \sum_r \sigma_r^2. \end{aligned}$	$\begin{aligned} \Phi &= \sum_r \sigma_r \mathbf{s}_r(\psi_r,) \\ \Phi^* \Phi = \mathbf{R} &= \sum_r \sigma_r^2(\psi_r,) \\ \mathrm{tr} \mathbf{R} &= \sum_r \sigma_r^2. \end{aligned}$	
penev_A_E00004 (2.4)	006	■ 0.303 ● W -1	$\begin{aligned} &\left(\hat{\mathbf{x}}, \hat{\mathbf{x}}'\right)_V \equiv V \delta_{\mathbf{x}, \mathbf{x}'} \\ &\left\langle \hat{t}, \hat{t}' \right\rangle_S \equiv T \delta_{t, t'} \end{aligned}$	$\begin{aligned} (\hat{\mathbf{x}}, \hat{\mathbf{x}}')_V &\equiv V \delta_{\mathbf{x}, \mathbf{x}'} \\ \langle \hat{t}, \hat{t}' \rangle_S &\equiv T \delta_{t, t'} \end{aligned}$	
penev_A_ E00075 (4.17)	018	■ 0.314 ● S +1	$\begin{aligned} \mathrm{0} &= \sum_f \mathrm{s_f}(\Psi_f,) \\ \mathrm{0}^* &= \sum_f \Psi_f \left\langle \mathrm{s_f}, \right\rangle. \end{aligned}$	$\begin{aligned} \mathbf{O} &= \sum_f \mathbf{s_f}(\Psi_f,) \\ \mathbf{O}^* &= \sum_f \Psi_f \langle \mathbf{s}_f, \rangle. \end{aligned}$	
penev_A_ E00030 (2.33)	007	■ 0.397 ● S -1	$\Phi^{*+} \Phi^* = \mathbf{P}_{\mathrm{range}(\Phi)} = \mathbf{P}' = \sum_{\sigma>0} \mathbf{s}_\sigma \left\langle \mathbf{s}_\sigma, \right\rangle.$	$\Phi^{*+} \Phi^* = \mathbf{P}_{\mathrm{range}(\Phi)} = \mathbf{P}' = \sum_{\sigma>0} \mathbf{s}_\sigma \langle \mathbf{s}_\sigma, \rangle.$	
penev_A_ E00039 (2.42)	008	■ 0.403 ● N +1	$\left\langle \Phi \psi_\sigma, \Phi \psi_{\sigma'} \right\rangle = \sigma \sigma' \frac{1}{T} \sum_t a_\sigma^t a_{\sigma'}^t = \sigma^2 \delta_{\sigma_1 \sigma'}$	$\langle \Phi \psi_\sigma, \Phi \psi_{\sigma'} \rangle = \sigma \sigma' \frac{1}{T} \sum_t a_\sigma^t a_{\sigma'}^t = \sigma^2 \delta_{\sigma, \sigma'}.$	
penev_A_ E00035 (2.48)	008	■ 0.409 ● W -2	$\left\ \mathbf{P}' \hat{t} \right\ _S^2 = \left\langle \mathbf{P}' \hat{t}, \mathbf{P}' \hat{t} \right\rangle = \sum_{\sigma>0} \left\langle \mathbf{s}_\sigma, \hat{t} \right\rangle^2 = \sum_{\sigma>0} u_\sigma^{t^2}.$	$\ \mathbf{P}' \hat{t} \ _S^2 = \langle \mathbf{P}' \hat{t}, \mathbf{P}' \hat{t} \rangle = \sum_{\sigma>0} \langle \mathbf{s}_\sigma, \hat{t} \rangle^2 = \sum_{\sigma>0} u_\sigma^{t^2}.$	
penev_A_E00007	019	■ 0.429 ● C +4	$\begin{aligned} K^{\{1\}} &= \mathrm{K} = \sum_{r=1}^N \psi_r \left(\frac{1}{\sigma_r} \right)^1 (\psi_r,) \\ K^{(0)} &= P = \sum_{r=1}^N \psi_r \left(\frac{1}{\sigma_r} \right)^0 (\psi_r,) \\ K^{(-1)} &= K^{(-1)} = \sum_{r=1}^N \psi_r \left(\frac{1}{\sigma_r} \right)^{-1} (\psi_r,) \end{aligned}$	$\begin{aligned} \mathbf{K}^{(1)} &= \mathbf{K} = \sum_{r=1}^N \psi_r \left(\frac{1}{\sigma_r} \right)^1 (\psi_r,) \\ \mathbf{K}^{(0)} &= \mathbf{P} = \sum_{r=1}^N \psi_r \left(\frac{1}{\sigma_r} \right)^0 (\psi_r,) \\ \mathbf{K}^{(-1)} &= \mathbf{K}^{(-1)} = \sum_{r=1}^N \psi_r \left(\frac{1}{\sigma_r} \right)^{-1} (\psi_r,) \end{aligned}$	
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5-0.9 ■ <0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured					

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penev_A_EQ0055 (3.6)	014	■ 0.496 ● W +2	$\begin{aligned} F_{\{r\}}^{\text{same}}(u) &= \mathcal{P}\left[D_r^{(p,q)} \leq u\right]_{(p,q) \in C_{\text{aame}}} \\ F_{\{r\}}^{\text{diff}}(u) &= \mathcal{P}\left[D_r^{(p,q)} \leq u\right]_{(p,q) \in C_{\text{diff}}} \end{aligned}$	$F_r^{\text{same}}(u) = \mathcal{P}\left[D_r^{(p,q)} \leq u\right]_{(p,q) \in C_{\text{aame}}}$ $F_r^{\text{diff}}(u) = \mathcal{P}\left[D_r^{(p,q)} \leq u\right]_{(p,q) \in C_{\text{diff}}}$	$F_r^{\text{same}}(u) = \mathcal{P}[D_r^{(p,q)} \leq u]_{(p,q) \in C_{\text{aame}}}$ $F_r^{\text{diff}}(u) = \mathcal{P}[D_r^{(p,q)} \leq u]_{(p,q) \in C_{\text{diff}}}$
penev_A_EQ0070 (4.12)	018	■ 0.502 ● N +0	$\mathbf{0}^+ \mathbf{0} = \mathbf{1}_U.$	$\mathbf{0}^+ \mathbf{0} = \mathbf{1}_U.$	$\mathbf{0}^+ \mathbf{0} = \mathbf{1}_U.$
penev_A_EQ0084 (4.26)	019	■ 0.525 ● C +3	$\begin{aligned} \mathrm{K} &= \mathrm{O}^* \Phi^{*+} = \sum_{\mathbf{f}} \Psi_{\mathbf{f}} \left(\frac{1}{\sigma_{\mathbf{f}}} \right)^1 (\Psi_{\mathbf{f}},) \\ 1 &= \mathrm{O}^* \mathrm{O} = \sum_{\mathbf{f}} \Psi_{\mathbf{f}} \left(\frac{1}{\sigma_{\mathbf{f}}} \right)^0 (\Psi_{\mathbf{f}},) \\ \mathrm{K}^{-1} &= \Phi^* \mathrm{O}^{*+} = \sum_{\mathbf{f}} \Psi_{\mathbf{f}} \left(\frac{1}{\sigma_{\mathbf{f}}} \right)^{-1} (\Psi_{\mathbf{f}},) \\ \mathrm{R} &= \Phi^* \Phi = \sum_{\mathbf{f}} \Psi_{\mathbf{f}} \left(\frac{1}{\sigma_{\mathbf{f}}} \right)^{-2} (\Psi_{\mathbf{f}},) \end{aligned}$	$\begin{aligned} \mathbf{K} &= \mathbf{O}^* \Phi^{*+} = \sum_{\mathbf{f}} \Psi_{\mathbf{f}} \left(\frac{1}{\sigma_{\mathbf{f}}} \right)^1 (\Psi_{\mathbf{f}},) \\ \mathbf{1} &= \mathbf{O}^* \mathbf{O} = \sum_{\mathbf{f}} \Psi_{\mathbf{f}} \left(\frac{1}{\sigma_{\mathbf{f}}} \right)^0 (\Psi_{\mathbf{f}},) \\ \mathbf{K}^{-1} &= \Phi^* \mathbf{O}^{*+} = \sum_{\mathbf{f}} \Psi_{\mathbf{f}} \left(\frac{1}{\sigma_{\mathbf{f}}} \right)^{-1} (\Psi_{\mathbf{f}},) \\ \mathbf{R} &= \Phi^* \Phi = \sum_{\mathbf{f}} \Psi_{\mathbf{f}} \left(\frac{1}{\sigma_{\mathbf{f}}} \right)^{-2} (\Psi_{\mathbf{f}},) \end{aligned}$	$\begin{aligned} \mathbf{K} &= \mathbf{O}^* \Phi^{*+} = \sum_{\mathbf{f}} \Psi_{\mathbf{f}} \left(\frac{1}{\sigma_{\mathbf{f}}} \right)^1 (\Psi_{\mathbf{f}},) \\ \mathbf{1} &= \mathbf{O}^* \mathbf{O} = \sum_{\mathbf{f}} \Psi_{\mathbf{f}} \left(\frac{1}{\sigma_{\mathbf{f}}} \right)^0 (\Psi_{\mathbf{f}},) \\ \mathbf{K}^{-1} &= \Phi^* \mathbf{O}^{*+} = \sum_{\mathbf{f}} \Psi_{\mathbf{f}} \left(\frac{1}{\sigma_{\mathbf{f}}} \right)^{-1} (\Psi_{\mathbf{f}},) \\ \mathbf{R} &= \Phi^* \Phi = \sum_{\mathbf{f}} \Psi_{\mathbf{f}} \left(\frac{1}{\sigma_{\mathbf{f}}} \right)^{-2} (\Psi_{\mathbf{f}},) \end{aligned}$
penev_A_EQ0021 (2.24)	007	■ 0.544 ● C +4	$\Phi^* \Phi \psi_{\sigma} \equiv \mathbf{R} \psi_{\sigma} = \sigma^2 \psi_{\sigma} \text{ with } (\psi_{\sigma}, \psi_{\sigma'}) = \delta_{\sigma, \sigma'}$	$\Phi^* \Phi \psi_{\sigma} \equiv \mathbf{R} \psi_{\sigma} = \sigma^2 \psi_{\sigma} \text{ with } (\psi_{\sigma}, \psi_{\sigma'}) = \delta_{\sigma, \sigma'}$	$\Phi^* \Phi \psi_{\sigma} \equiv \mathbf{R} \psi_{\sigma} = \sigma^2 \psi_{\sigma} \text{ with } (\psi_{\sigma}, \psi_{\sigma'}) = \delta_{\sigma, \sigma'}$
penev_A_EQ0094 (4.35)	020	■ 0.552 ● N +2	$\phi = \mathbf{P}_N \phi = \Phi^* \Phi^{*+} \phi = \Phi^* \mathbf{O}^{*+} O = \mathbf{K}^{(-1)} O = \sum_{r=1}^N a_r \sigma_r \psi_r$	$\phi = \mathbf{P}_N \phi = \Phi^* \Phi^{*+} \phi = \Phi^* \mathbf{O}^{*+} O = \mathbf{K}^{(-1)} O = \sum_{r=1}^N a_r \sigma_r \psi_r.$	$\phi = \mathbf{P}_N \phi = \Phi^* \Phi^{*+} \phi = \Phi^* \mathbf{O}^{*+} O = \mathbf{K}^{(-1)} O = \sum_{r=1}^N a_r \sigma_r \psi_r.$
penev_A_EQ0057 (3.8)	015	■ 0.569 ● C -3	$w_t^{Fisher} \propto \sum_{r=1}^N (m_r^{sqame} - m_r^{diff}) (H^{intmg})_{rt}^{-1}$	$w_t^{Fisher} \propto \sum_{r=1}^N (m_r^{sqame} - m_r^{diff}) (H^{intmg})_{rt}^{-1}$	$w_t^{Fisher} \propto \sum_{r=1}^N (m_r^{sqame} - m_r^{diff}) (H^{intmg})_{rt}^{-1}$
penev_A_EQ0069 (4.11)	018	■ 0.571 ● N +2	$(\hat{\mathbf{x}}, \hat{\mathbf{y}}) = V \delta_{\mathbf{x}, \mathbf{y}} = \frac{1}{T} \sum_t O^t(\mathbf{x}) O^t(\mathbf{y}) = \langle \mathbf{O} \hat{\mathbf{x}}, \mathbf{O} \hat{\mathbf{y}} \rangle = (\hat{\mathbf{x}}, \mathbf{O}^* \mathbf{O} \hat{\mathbf{y}})$	$(\hat{\mathbf{x}}, \hat{\mathbf{y}}) = V \delta_{\mathbf{x}, \mathbf{y}} = \frac{1}{T} \sum_t O^t(\mathbf{x}) O^t(\mathbf{y}) = \langle \mathbf{O} \hat{\mathbf{x}}, \mathbf{O} \hat{\mathbf{y}} \rangle = (\hat{\mathbf{x}}, \mathbf{O}^* \mathbf{O} \hat{\mathbf{y}})$	$(\hat{\mathbf{x}}, \hat{\mathbf{y}}) = V \delta_{\mathbf{x}, \mathbf{y}} = \frac{1}{T} \sum_t O^t(\mathbf{x}) O^t(\mathbf{y}) = \langle \mathbf{O} \hat{\mathbf{x}}, \mathbf{O} \hat{\mathbf{y}} \rangle = (\hat{\mathbf{x}}, \mathbf{O}^* \mathbf{O} \hat{\mathbf{y}})$
penev_A_EQ0111 (4.46)	022	■ 0.585 ● N +1	$\mathbf{q}_m \equiv \mathbf{q}_{\mathbf{x}_m} = \mathbf{O} \hat{\mathbf{x}}_m.$	$\mathbf{q}_m \equiv \mathbf{q}_{\mathbf{x}_m} = \mathbf{O} \hat{\mathbf{x}}_m.$	$\mathbf{q}_m \equiv \mathbf{q}_{\mathbf{x}_m} = \mathbf{O} \hat{\mathbf{x}}_m.$
penev_A_EQ0061 (4.3)	017	■ 0.590 ● N +0	$\Psi_{\mathbf{f}}(\mathbf{x}) = e^{\mathbf{if} \cdot \mathbf{x}}.$	$\Psi_{\mathbf{f}}(\mathbf{x}) = e^{\mathbf{if} \cdot \mathbf{x}}.$	$\Psi_{\mathbf{f}}(\mathbf{x}) = e^{\mathbf{if} \cdot \mathbf{x}}.$
penev_A_EQ0106 (4.47)	022	■ 0.622 ● C +4	$(\hat{\mathbf{x}}, \mathbf{O}^* \mathbf{q}_{\mathbf{x}_m}) = (\hat{\mathbf{x}}, \mathbf{O}^* \mathbf{O} \hat{\mathbf{x}}_m) = (\hat{\mathbf{x}}, \mathbf{P} \hat{\mathbf{x}}_m) \equiv (\hat{\mathbf{x}}, \mathbf{P}_{\mathbf{x}_m}) \equiv P_{\mathbf{x}_m}(\mathbf{x}).$	$(\hat{\mathbf{x}}, \mathbf{O}^* \mathbf{q}_{\mathbf{x}_m}) = (\hat{\mathbf{x}}, \mathbf{O}^* \mathbf{O} \hat{\mathbf{x}}_m) = (\hat{\mathbf{x}}, \mathbf{P} \hat{\mathbf{x}}_m) \equiv (\hat{\mathbf{x}}, \mathbf{P}_{\mathbf{x}_m}) \equiv P_{\mathbf{x}_m}(\mathbf{x}).$	$(\hat{\mathbf{x}}, \mathbf{O}^* \mathbf{q}_{\mathbf{x}_m}) = (\hat{\mathbf{x}}, \mathbf{O}^* \mathbf{O} \hat{\mathbf{x}}_m) = (\hat{\mathbf{x}}, \mathbf{P} \hat{\mathbf{x}}_m) \equiv (\hat{\mathbf{x}}, \mathbf{P}_{\mathbf{x}_m}) \equiv P_{\mathbf{x}_m}(\mathbf{x}).$
Conf. MathPix confidence: ■ ≥ 0.9 ■ 0.5–0.9 ■ < 0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured					

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penev_A_ EQ0018 (2.21)	007	■ 0.644 ● N +1	$\left(\mathrm{u}^{\prime},\Phi^{*}\Phi\mathrm{u}\right)=\left\langle\Phi\mathrm{u}^{\prime},\Phi\mathrm{u}\right\rangle=\left(\Phi^{*}\Phi\mathrm{u}^{\prime},\mathrm{u}\right).$ $\left(\mathrm{u}^{\prime},\Phi^{*}\Phi\mathrm{u}\right)=\left\langle\Phi\mathrm{u}^{\prime},\Phi\mathrm{u}\right\rangle=\left(\Phi^{*}\Phi\mathrm{u}^{\prime},\mathrm{u}\right).$		
penev_A_ EQ0077 (4.19)	018	■ 0.654 ● S +1	$\mathrm{K}=\sum_{\mathbf{f}}\Psi_{\mathbf{f}}\frac{1}{\sigma_{\mathbf{f}}}(\Psi_{\mathbf{f}},)=0^{*}\Phi^{*+}$	$\mathbf{K}=\sum_{\mathbf{f}}\Psi_{\mathbf{f}}\frac{1}{\sigma_{\mathbf{f}}}(\Psi_{\mathbf{f}},)=0^{*}\Phi^{*+}$	$\mathbf{K}=\sum_{\mathbf{f}}\Psi_{\mathbf{f}}\frac{1}{\sigma_{\mathbf{f}}}(\Psi_{\mathbf{f}},)=\mathbf{O}^{*}\Phi^{*+}.$
penev_A_ EQ0092 (4.33)	020	■ 0.657 ● N +0	$0^{\{*\}}\ 0=\Phi^{\{*\}}\ \phi$	$O^{*+}O=\Phi^{*+}\phi$	$\mathbf{O}^{*+}O=\Phi^{*+}\phi$
penev_A_ EQ0096 (4.37)	021	■ 0.658 ● S +0	$0_{\{m\}}=\left\langle\mathrm{q}_{\mathbf{x}_{\{m\}}},\mathrm{s}\right\rangle$	$O_m=\langle\mathbf{q}_{\mathbf{x}_m},\mathbf{s}\rangle$	$O_m=\langle\mathbf{q}_{\mathbf{x}_m},\mathfrak{s}\rangle$
penev_A_EQ0063 (4.5)	017	■ 0.666 ● N +0	$\Phi^{*}\Phi=\mathbf{R}=\sum_{\mathbf{f}}\Psi_{\mathbf{f}}\sigma_{\mathbf{f}}^2(\Psi_{\mathbf{f}},)$	$\Phi^{*}\Phi=\mathbf{R}=\sum_{\mathbf{f}}\Psi_{\mathbf{f}}\sigma_{\mathbf{f}}^2(\Psi_{\mathbf{f}},)$	$\Phi^{*}\Phi=\mathbf{R}=\sum_{\mathbf{f}}\Psi_{\mathbf{f}}\sigma_{\mathbf{f}}^2(\Psi_{\mathbf{f}},)$
penev_A_ EQ0112 (4.53)	022	■ 0.692 ● W -1	$\mathbf{p}_l=\sum_{n=1}^{ M }\mathbf{B}_{ln}\mathbf{q}_n.$	$\mathbf{p}_l=\sum_{n=1}^{ M }\mathbf{B}_{ln}\mathbf{q}_n.$	$\mathbf{p}_l=\sum_{n=1}^{ M }\mathbf{B}_{ln}\mathbf{q}_n.$
penev_A_ EQ0026 (2.29)	007	■ 0.693 ● N +1	$\left\langle\mathbf{s}_{\sigma},\mathbf{s}_{\sigma'}\right\rangle=\delta_{\sigma,\sigma'}$	$\langle\mathbf{s}_{\sigma},\mathbf{s}_{\sigma'}\rangle=\delta_{\sigma,\sigma'}$	$\langle\mathfrak{s}_{\sigma},\mathfrak{s}_{\sigma'}\rangle=\delta_{\sigma,\sigma'}$
penev_A_ EQ0113 (4.54)	022	■ 0.695 ● S -2	$\langle\mathbf{q}_m,\mathbf{p}_i\rangle=\delta_{m,i}.$	$\langle\mathbf{q}_m,\mathbf{p}_i\rangle=\delta_{m,i}.$	$\langle\mathbf{q}_m,\mathbf{p}_i\rangle=\delta_{m,i}.$
penev_A_ EQ0071 (4.13)	018	■ 0.710 ● S +1	$O^{*}O=\sum_f\Psi_f(\Psi_f,)$	$O^{*}O=\sum_f\Psi_f(\Psi_f,)$	$\mathbf{O}^{*}\mathbf{O}=\sum_{\mathbf{f}}\Psi_{\mathbf{f}}(\Psi_{\mathbf{f}},).$
penev_A_ EQ0031 (2.34)	007	■ 0.715 ● N +0	$\Phi^{*}\Phi^{*+}=\mathbf{P}_{\mathrm{range}}\left(\Phi^{*}\right)=\mathbf{P}=\sum_{\sigma>0}\psi_{\sigma}(\psi_{\sigma},).$	$\Phi^{*}\Phi^{*+}=\mathbf{P}_{\mathrm{range}}\left(\Phi^{*}\right)=\mathbf{P}=\sum_{\sigma>0}\psi_{\sigma}(\psi_{\sigma},).$	$\Phi^{*}\Phi^{*+}=\mathbf{P}_{\mathrm{range}}(\Phi^{*})=\mathbf{P}=\sum_{\sigma>0}\psi_{\sigma}(\psi_{\sigma},).$
penev_A_ EQ0028 (2.31)	007	■ 0.721 ● N +1	$\Phi^{*}=\sum_{\sigma>0}\sigma\psi_{\sigma}\langle\mathbf{s}_{\sigma},\rangle.$	$\Phi^{*}=\sum_{\sigma>0}\sigma\psi_{\sigma}\langle\mathbf{s}_{\sigma},\rangle.$	$\Phi^{*}=\sum_{\sigma>0}\sigma\psi_{\sigma}(\mathfrak{s}_{\sigma},).$
penev_A_ EQ0107 (4.48)	022	■ 0.724 ● S -2	$O^{rec}(\mathbf{x})=O_ma_{\mathbf{x}_{\mathbf{wa}}}(\mathbf{x})$	$O^{rec}(\mathbf{x})=O_ma_{\mathbf{x}_{\mathbf{wa}}}(\mathbf{x})$	$O^{rec}(\mathbf{x})=O_ma_{\mathbf{x}_m}(\mathbf{x})$
penev_A_ EQ0040 (2.43)	008	■ 0.742 ● W +0	$\mathcal{P}\left[\phi^t\right]=\mathcal{P}\left[\mathbf{P}'\hat{t}\right]=\prod_{\sigma>0}\mathcal{P}\left[a_{\sigma}^t\right].$	$\mathcal{P}\left[\phi^t\right]=\mathcal{P}\left[\mathbf{P}'\hat{t}\right]=\prod_{\sigma>0}\mathcal{P}\left[a_{\sigma}^t\right].$	$\mathcal{P}[\phi^t]=\mathcal{P}[\mathbf{P}'\hat{t}]=\prod_{\sigma>0}\mathcal{P}[a_{\sigma}^t].$
penev_A_ EQ0100 (4.41)	022	■ 0.748 ● N -2	$\langle q_{x_m},p_{x_m}\rangle=1$	$\langle q_{x_m},p_{x_m}\rangle=1$	$\langle\mathbf{q}_{\mathbf{x}_m},\mathbf{p}_{\mathbf{x}_m}\rangle=1$
penev_A_ EQ0073 (4.15)	018	■ 0.789 ● N +0	$O^{*}o_{\mathbf{f}}=\Psi_f.$	$O^{*}o_{\mathbf{f}}=\Psi_f.$	$\mathbf{O}^{*}\mathbf{o}_{\mathbf{f}}=\Psi_{\mathbf{f}}.$
penev_A_ EQ0104 (4.45)	022	■ 0.789 ● N -2	$O^{rec}(\mathbf{x})=\left(\hat{\mathbf{x}},\mathbf{O}^{*}\mathbf{s}^{rec}\right).$	$O^{rec}(\mathbf{x})=\left(\hat{\mathbf{x}},\mathbf{O}^{*}\mathbf{s}^{rec}\right).$	$O^{rec}(\mathbf{x})=\left(\hat{\mathbf{x}},\mathbf{O}^{*}\mathfrak{s}^{rec}\right).$
penev_A_ EQ0047 (2.50)	008	■ 0.799 ● N +0	$\phi_N^{\mathrm{rec}}=(\Phi_N)^*(\Phi_N)^{*+}\phi=\mathbf{P}_N\phi=\sum_{r=1}^Na_r\sigma_r\psi_r$	$\phi_N^{\mathrm{rec}}=(\Phi_N)^*(\Phi_N)^{*+}\phi=\mathbf{P}_N\phi=\sum_{r=1}^Na_r\sigma_r\psi_r$	$\phi_N^{\mathrm{rec}}=(\Phi_N)^*(\Phi_N)^{*+}\phi=\mathbf{P}_N\phi=\sum_{r=1}^Na_r\sigma_r\psi_r$
penev_A_ EQ0034 (2.37)	008	■ 0.801 ● W -2	$\mathbf{P}'\hat{t}=\sum_{\sigma>0}\mathbf{s}_{\sigma}\langle\mathbf{s}_{\sigma},\hat{t}\rangle\equiv\sum_{\sigma>0}a_{\sigma}^i\mathbf{s}_{\sigma}$	$\mathbf{P}'\hat{t}=\sum_{\sigma>0}\mathbf{s}_{\sigma}\langle\mathbf{s}_{\sigma},\hat{t}\rangle\equiv\sum_{\sigma>0}a_{\sigma}^i\mathbf{s}_{\sigma}$	$\mathbf{P}'\hat{t}=\sum_{\sigma>0}\mathfrak{s}_{\sigma}(\mathfrak{s}_{\sigma},\hat{t})\equiv\sum_{\sigma>0}a_{\sigma}^t\mathfrak{s}_{\sigma}$
penev_A_ EQ0074 (4.16)	018	■ 0.811 ● N +0	$o_f=s_f.$	$o_f=s_f.$	$\mathbf{o}_{\mathbf{f}}=\mathfrak{s}_{\mathbf{f}}.$

Conf. MathPix confidence: ■ ≥ 0.9 ■ 0.5–0.9 ■ < 0.5 — no value
Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured

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penev_A_ EQ0079 (4.21)	018	■ 0.822 ● C +3	$\begin{aligned} \backslash begin{aligned} \backslash left\backslash\Phi^{\wedge\{*\}} \backslash phi\right\backslash_{\{S\}^{\wedge\{2\}}= & \backslash left\backslash\langle\Phi^{\wedge\{*\}} \backslash phi, \backslash\Phi^{\wedge\{*\}} \backslash phi\right\backslash\langle= \\ \backslash\langle\mathrm{\mathbf{0} \ K} \backslash phi, \mathrm{\mathbf{0} \ K} \backslash phi\rangle= \backslash & \backslash\left(\mathrm{\mathbf{K}} \backslash phi, \mathrm{\mathbf{0}}^{\wedge\{*\}} \mathrm{\mathbf{0} \ K} \right. \\ \backslash phi\right)= & \backslash(\mathrm{\mathbf{K}} \backslash phi, \mathrm{\mathbf{0}} \mathrm{\mathbf{K}} \backslash phi)=\backslash \\ \mathrm{\mathbf{K}} \backslash phi\backslash_{\{U\}^{\wedge\{2\}} \ . \end{aligned}$	$\begin{aligned} \left\ \Phi^{*+}\phi\right\ _S^2 &= \left\langle \Phi^{*+}\phi, \Phi^{*+}\phi \right\rangle = \left\langle \mathbf{O}\mathbf{K}\phi, \mathbf{O}\mathbf{K}\phi \right\rangle = \\ &= (\mathbf{K}\phi, \mathbf{O}^*\mathbf{O}\mathbf{K}\phi) = (\mathbf{K}\phi, \mathbf{K}\phi) = \left\ \mathbf{K}\phi\right\ _U^2. \end{aligned}$	$\begin{aligned} \left\ \Phi^{*+}\phi\right\ _S^2 &= \left\langle \Phi^{*+}\phi, \Phi^{*+}\phi \right\rangle = \left\langle \mathbf{O}\mathbf{K}\phi, \mathbf{O}\mathbf{K}\phi \right\rangle = \\ &= (\mathbf{K}\phi, \mathbf{O}^*\mathbf{O}\mathbf{K}\phi) = \left(\mathbf{K}\phi, \mathbf{K}\phi\right) = \left\ \mathbf{K}\phi\right\ _U^2. \end{aligned}$
penev_A_ EQ0108 (4.49)	022	■ 0.827 ● S -2	$a_{\mathrm{\mathbf{x}}_m}(\mathrm{\mathbf{x}})_{\left(\mathrm{\mathbf{x}}\right)}=\frac{P_{\mathrm{\mathbf{x}}_m}(\mathrm{\mathbf{x}})}{P_{\mathrm{\mathbf{x}}_m}(\mathrm{\mathbf{x}}_m)}\left(\mathrm{\mathbf{x}}\right)$	$a_{\mathbf{x}_m}(\mathbf{x})=\frac{P_{\mathbf{x}_m}(\mathbf{x})}{P_{\mathbf{x}_m}(\mathbf{x}_m)}.$	$a_{\mathbf{x}_m}(\mathbf{x})=\frac{P_{\mathbf{x}_m}(\mathbf{x})}{P_{\mathbf{x}_m}(\mathbf{x}_m)}.$
penev_A_EQ0066 (4.8)	018	■ 0.844 ● N +0	$K\,\Phi^*=O^*$	$K\Phi^*=O^*$	$\mathbf{K}\Phi^*=\mathbf{O}^*$
penev_A_ EQ0024 (2.27)	007	■ 0.852 ● N +2	$\mathrm{\mathbf{R}}=\mathrm{\mathbf{R}}\cdot\mathrm{\mathbf{P}}=\sum_{\sigma>0}\sigma^2\psi_{\sigma}(\psi_{\sigma},)\,\,\,\backslash\sigma^{\wedge\{2\}}\,\,\,\backslash\psi_{\{\sigma\}}\backslash\left(\backslash\psi_{\{\sigma\}},\backslash\right)\,\,\,$	$\mathbf{R}=\mathbf{R}\cdot\mathbf{P}=\sum_{\sigma>0}\sigma^2\psi_{\sigma}(\psi_{\sigma},).$	$\mathbf{R}=\mathbf{R}\mathbf{P}=\sum_{\sigma>0}\sigma^2\psi_{\sigma}(\psi_{\sigma},).$
penev_A_EQ0056 (3.7)	014	■ 0.852 ● S +0	$\mu_2=\left\langle\left(\mathbf{u}-\mathbf{v}\right)^2\right\rangle\right\r$		

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penev_A_ EQ0072 (4.14)	018	■ 0.911 ● N +2	$\backslash\mathrm{bf}\{0\} \ \backsi_{\back\mathrm{bf}\{f\}}=\back\mathrm{bf}\{o_{\back\mathrm{bf}\{f\}} \ \in S \ .$	$\mathbf{O}\Psi_{\mathbf{f}}=\mathbf{o}_{\mathbf{f}}\in S.$	$\mathbf{O}\Psi_{\mathbf{f}}=\mathbf{o}_{\mathbf{f}}\in S.$
penev_A_EQ0051 (3.2)	014	■ 0.914 ● S -2	$T \times T=\back\{ (p,p) \}_{p \in T} \cup C_{\text{same}} \cup C_{\text{diff}} \ .$	$T \times T = \{ (p,p) \}_{p \in T} \cup C_{\text{same}} \cup C_{\text{diff}} .$	$T \times T = \{ (p,p) \}_{p \in T} \cup C_{\text{same}} \cup C_{\text{diff}} .$
penev_A_ EQ0090 (4.31)	020	■ 0.925 ● N +1	$0=\back\mathrm{bf}\{K\} \ \back\phi=\mathrm{bf}\{0\}^{\back\ast} \ \back\Phi^{\back\ast+} \ \back\phi=\back\sum_{r=1}^{\back N} \ a_r \ \back\psi_r$	$O=\mathbf{K}\phi=\mathbf{O}^*\Phi^{*+}\phi=\sum_{r=1}^Na_r\psi_r$	$O=\mathbf{K}\phi=\mathbf{O}^*\Phi^{*+}\phi=\sum_{r=1}^Na_r\psi_r$
penev_A_ EQ0085 (4.27)	019	■ 0.928 ● W +1	$\back\mathrm{bf}\{K\}^{\back n}=\back\sum_{\back\mathrm{bf}\{f\}} \ \back\psi_{\back\mathrm{bf}\{f\}}\back\left(\frac{1}{\back\sigma_{\back\mathrm{bf}\{f\}}}\right)^{\back n}\back\left(\back\psi_{\back\mathrm{bf}\{f\}}\right) \ .$	$\mathbf{K}^n=\sum_{\mathbf{f}}\Psi_{\mathbf{f}}\left(\frac{1}{\sigma_{\mathbf{f}}}\right)^n(\Psi_{\mathbf{f}},).$	$\mathbf{K}^n=\sum_{\mathbf{f}}\Psi_{\mathbf{f}}\left(\frac{1}{\sigma_{\mathbf{f}}}\right)^n(\Psi_{\mathbf{f}},).$
penev_A_ EQ0097 (4.38)	021	■ 0.929 ● C +3	$\back\mathrm{bf}\{q_{\back\mathrm{bf}\{x\}_{\back m}}\} \ \equiv \back\mathrm{bf}\{0\} \ \hat{\back\mathrm{bf}\{x\}_{\back m}} \ .$	$\mathbf{q}_{\mathbf{x}_m}\equiv\mathbf{O}\hat{\mathbf{x}}_m.$	$\mathbf{q}_{\mathbf{x}_m}\equiv\mathbf{O}\hat{\mathbf{x}}_m.$
penev_A_ EQ0032 (2.35)	007	■ 0.932 ● W -2	$\back\mathrm{cal}\{P\}\back\left[\back\Phi^{\back\ast} \ \back\mathrm{bf}\{s\}\right] \ \propto \back\exp \ \back\left\{ -\frac{1}{2} \ \back\ \mathbf{P}'\mathbf{s}\ _S^2 \right\} \ .$	$\mathcal{P}\left[\Phi^*\mathbf{s}\right]\propto\exp\left\{-\frac{1}{2}\left\ \mathbf{P}'\mathbf{s}\right\ _S^2\right\}.$	$\mathcal{P}[\Phi^*\mathbf{s}]\propto\exp\left\{-\frac{1}{2}\left\ \mathbf{P}'\mathbf{s}\right\ _S^2\right\}.$
penev_A_ EQ0042 (2.45)	008	■ 0.934 ● C +5	$\back\begin{aligned} \back\Phi_{\back N} \ \&= \back\sum_{r=1}^{\back N} \ \back\sigma_r \ \back\left(\back\psi_r, \right) \\ \back\mathrm{bf}\{s_{\back r}\}\back\left(\back\psi_{\back r},\back\right) \ \&\back\left(\back\Phi_{\back N}\back\right)^{\back\ast} \ \back\Phi_{\back N}=\back\mathrm{bf}\{R\}_{\back N} \ \&= \\ \back\sum_{r=1}^{\back N} \ \back\sigma_r \ \back\{ \ }^{\back 2}\back\left(\back\psi_{\back r},\back\right) \ \& \ \back\operatorname{tr} \ \back\mathrm{bf}\{R\}_{\back N} \ \&= \back\sum_{r=1}^{\back N} \ \back\sigma_r \ \{ \ }^{\back 2} \ . \end{aligned}$	$\begin{aligned} \Phi_N &= \sum_{r=1}^N \sigma_r \mathbf{s}_r \left(\psi_r, \right) \\ (\Phi_N)^* \Phi_N &= \mathbf{R}_N = \sum_{r=1}^N \sigma_r^2 \left(\psi_r, \right) \\ \mathrm{tr} \mathbf{R}_N &= \sum_{r=1}^N \sigma_r^2. \end{aligned}$	$\begin{aligned} \Phi_N &= \sum_{r=1}^N \sigma_r \mathbf{s}_r \left(\psi_r, \right) \\ (\Phi_N)^* \Phi_N &= \mathbf{R}_N = \sum_{r=1}^N \sigma_r^2 \left(\psi_r, \right) \\ \mathrm{tr} \mathbf{R}_N &= \sum_{r=1}^N \sigma_r^2. \end{aligned}$
penev_A_ EQ0101 (4.42)	022	■ 0.945 ● N +0	$\back\mathrm{bf}\{s_{\back }\}=\back\mathrm{bf}\{Q_{\back\mathrm{bf}\{x\}_{\back m}}\} \ \back\mathrm{bf}\{s\}=\back\mathrm{bf}\{p_{\back\mathrm{bf}\{x\}_{\back m}}\}\back\left\back\angle \back\mathrm{bf}\{q_{\back\mathrm{bf}\{x\}_{\back m}}\}, \ \back\mathrm{bf}\{s\}\right\angle=0_{\back m} \ \back\mathrm{bf}\{p_{\back\mathrm{bf}\{x\}_{\back m}}\}$	$\mathbf{s}_{\parallel}=\mathbf{Q}_{\mathbf{x}_m}\mathbf{s}=\mathbf{p}_{\mathbf{x}_m}\langle\mathbf{q}_{\mathbf{x}_m},\mathbf{s}\rangle=O_m\mathbf{p}_{\mathbf{x}_m}$	$\mathbf{s}_{\parallel}=\mathbf{Q}_{\mathbf{x}_m}\mathbf{s}=\mathbf{p}_{\mathbf{x}_m}\langle\mathbf{q}_{\mathbf{x}_m},\mathbf{s}\rangle=O_m\mathbf{p}_{\mathbf{x}_m}$
penev_A_ EQ0099 (4.40)	022	■ 0.948 ● S -2	$\back\mathrm{bf}\{p_{\back x_{\back m}}\}=\frac{\back\mathrm{bf}\{q_{\back x_{\back m}}\}}{\back\left\back\angle \back\mathrm{bf}\{q_{\back x_{\back m}}\}, \ \back\mathrm{bf}\{q_{\back x_{\back m}}\}\right\angle} \ .$	$\mathbf{p}_{x_m}=\frac{\mathbf{q}_{x_m}}{\langle\mathbf{q}_{x_m},\mathbf{q}_{x_m}\rangle}.$	$\mathbf{p}_{\mathbf{x}_m}=\frac{\mathbf{q}_{\mathbf{x}_m}}{\langle\mathbf{q}_{\mathbf{x}_m},\mathbf{q}_{\mathbf{x}_m}\rangle}.$
penev_A_ EQ0115 (4.56)	022	■ 0.949 ● S +1	$\back\left\back\angle \back\mathrm{bf}\{q_{\back m}, \ \back\mathrm{bf}\{s_{\back }\}\right\angle=\back\sum_{l=1}^{\back M } \ 0_{\back l}\back\left\back\angle \back\mathrm{bf}\{q_{\back m}, \ \back\mathrm{bf}\{p_{\back l}\}\right\angle=\back\sum_{l=1}^{\back M } \ \back\mid 0_{\back l} \ \back\delta_{ml}=0_{\back m}=\back\left\back\angle \back\mathrm{bf}\{q_{\back m}, \ \back\mathrm{bf}\{s_{\back }\}\right\angle$	$\langle\mathbf{q}_m,\mathbf{s}_{\parallel}\rangle=\sum_{l=1}^{ \mathcal{M} }O_l\langle\mathbf{q}_m,\mathbf{p}_l\rangle=\sum_{l=1}^{ \mathcal{M} }O_l\delta_{ml}=O_m=\langle\mathbf{q}_m,\mathbf{s}\rangle$	$\langle\mathbf{q}_m,\mathbf{s}_{\parallel}\rangle=\sum_{l=1}^{ \mathcal{M} }O_l\langle\mathbf{q}_m,\mathbf{p}_l\rangle=\sum_{l=1}^{ \mathcal{M} }O_l\delta_{ml}=O_m=\langle\mathbf{q}_m,\mathbf{s}\rangle$
penev_A_ EQ0098 (4.39)	022	■ 0.960 ● C -5	$\back\mathrm{bf}\{Q_{\back\mathrm{bf}\{x\}_{\back m}}\}=\frac{\back\mathrm{bf}\{q_{\back\mathrm{bf}\{x\}_{\back m}}\}\back\left\back\angle \back\mathrm{bf}\{q_{\back\mathrm{bf}\{x\}_{\back m}}\}, \ \back\right\angle}{\back\left\back\angle \back\mathrm{bf}\{q_{\back\mathrm{bf}\{x\}_{\back m}}\}, \ \back\mathrm{bf}\{p_{\back\mathrm{bf}\{x\}_{\back m}}\}\right\angle}=\back\mathrm{bf}\{p_{\back\mathrm{bf}\{x\}_{\back m}}\}\back\left\back\angle \back\mathrm{bf}\{q_{\back\mathrm{bf}\{x\}_{\back m}}\}, \ \back\right\angle$	$\mathbf{Q}_{\mathbf{x}_m}=\frac{\mathbf{q}_{\mathbf{x}_m}\langle\mathbf{q}_{\mathbf{x}_m},\rangle}{\langle\mathbf{q}_{\mathbf{x}_m},\mathbf{q}_{\mathbf{x}_m}\rangle}=\mathbf{p}_{\mathbf{x}_m}\langle\mathbf{q}_{\mathbf{x}_m},\rangle$	$\mathbf{Q}_{\mathbf{x}_m}=\frac{\mathbf{q}_{\mathbf{x}_m}\langle\mathbf{q}_{\mathbf{x}_m},\rangle}{\langle\mathbf{q}_{\mathbf{x}_m},\mathbf{q}_{\mathbf{x}_m}\rangle}=\mathbf{p}_{\mathbf{x}_m}\langle\mathbf{q}_{\mathbf{x}_m},\rangle$
penev_A_ EQ0095 (4.36)	021	■ 0.968 ● W +1	$0(\back\mathrm{bf}\{x\})=\back\sum_{m=1}^{\back M } \ 0_{\back m} \ a_{\back m}(\back\mathrm{bf}\{x\})$	$O(\mathbf{x})=\sum_{m=1}^{ \mathcal{M} }O_ma_m(\mathbf{x})$	$O(\mathbf{x})=\sum_{m=1}^{ \mathcal{M} }O_ma_m(\mathbf{x})$
penev_A_ EQ0049 (2.52)	008	■ 0.975 ● N +0	$\back\left\back\left\back\left\back\phi_{\back N}^{\back r e c}\right\right\}_U^2=\left(\phi_{\back N}^{\back r e c},\phi_{\back N}^{\back r e c}\right)=\back\sum_{r=1}^{\back N} \ \sigma^2 a_r^2$	$\ \phi_N^{rec}\ _U^2=(\phi_N^{rec},\phi_N^{rec})=\sum_{r=1}^N\sigma^2a_r^2$	$\ \phi_N^{rec}\ _U^2=(\phi_N^{rec},\phi_N^{rec})=\sum_{r=1}^N\sigma^2a_r^2$
penev_A_ EQ0044 (2.47)	008	■ 0.976 ● N +0	$a_{\back r}(\back\phi) \ \equiv \ \sigma_{\back r}^{-1} \ (\back\psi_r, \back\phi)$	$a_r(\phi)\equiv\sigma_r^{-1}(\psi_r,\phi)$	$a_r(\phi)\equiv\sigma_r^{-1}(\psi_r,\phi)$
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured					

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
penev_A_ EQ0078 (4.20)	018	0.977 N +2	$\mathrm{OK}=\mathrm{OO}^*\Phi^{*+}=\mathrm{P}'\Phi^{*+}=\Phi^{*+}\Phi^{*+}=\Phi^{*+}$	$\mathrm{OK}=\mathrm{OO}^*\Phi^{*+}=\mathrm{P}'\Phi^{*+}=\Phi^{*+}$	
penev_A_EQ0008 (2.9)	006	0.977 C +5	$\begin{aligned} \ \phi\ _U^2 &= \langle \phi, \phi \rangle = \frac{1}{V} \sum_{\mathbf{x}} \phi(\mathbf{x}) \phi(\mathbf{x}) \\ \ \mathbf{s}\ _S^2 &= \langle \mathbf{s}, \mathbf{s} \rangle = \frac{1}{T} \sum_t s(t) s(t) \end{aligned}$	$\ \phi\ _U^2 = \langle \phi, \phi \rangle = \frac{1}{V} \sum_{\mathbf{x}} \phi(\mathbf{x}) \phi(\mathbf{x})$ $\ \mathbf{s}\ _S^2 = \langle \mathbf{s}, \mathbf{s} \rangle = \frac{1}{T} \sum_t s(t) s(t).$	
penev_A_ EQ0088 (4.29)	020	0.980 N +0	$\mathbf{K}^{(-2)} = \mathbf{R} = \sum_{r=1}^N \psi_r \left(\frac{1}{\sigma_r} \right)^{-2} (\psi_r,)$	$\mathbf{K}^{(-2)} = \mathbf{R} = \sum_{r=1}^N \psi_r \left(\frac{1}{\sigma_r} \right)^{-2} (\psi_r,)$	
penev_A_ EQ0081 (4.23)	018	0.981 W +1	$\mathrm{O}^{*+}O = OO = \mathrm{OK}\phi = \Phi^{*+}\phi.$	$\mathrm{O}^{*+}O = OO = \mathrm{OK}\phi = \Phi^{*+}\phi.$	
penev_A_ EQ0076 (4.18)	018	0.981 N +1	$O\,O^* = \sum_f s_f(s_f,)=P'.$	$\mathrm{OO}^* = \sum_{\mathbf{f}} s_{\mathbf{f}}(s_{\mathbf{f}},) = \mathbf{P}'.$	
penev_A_EQ0001 (2.1)	006	0.982 K +0	$R(\mathbf{x},\mathbf{y}) = \langle \phi^t(\mathbf{x}) \phi^t(\mathbf{y}) \rangle_T$	$R(\mathbf{x},\mathbf{y}) = \langle \phi^t(\mathbf{x}) \phi^t(\mathbf{y}) \rangle_T$	
penev_A_ EQ0020 (2.23)	007	0.986 N +0	$\ker \Phi^* \Phi = \ker \Phi.$	$\ker \Phi^* \Phi = \ker \Phi.$	
penev_A_EQ0052 (3.3)	014	0.991 N +0	$\mathrm{d}^{(p,q)} = (\Phi_N)^{*+} \phi^p \quad (\Phi_N)^{*+} \phi^q \in S$	$\mathbf{d}^{(p,q)} = (\Phi_N)^{*+} \phi^p \quad (\Phi_N)^{*+} \phi^q \in S$	
penev_A_ EQ0009 (2.10)	006	0.995 N +0	$(\phi,\phi') = \frac{1}{V} \sum_{\mathbf{x}} \phi(\mathbf{x}) \phi'(\mathbf{x})$	$(\phi,\phi') = \frac{1}{V} \sum_{\mathbf{x}} \phi(\mathbf{x}) \phi'(\mathbf{x})$	
penev_A_EQ0050 (3.1)	013	0.995 N +0	$\epsilon = \ \mathbf{P}_N \phi^k - \mathbf{P}_N \phi\ _U$	$\epsilon = \ \mathbf{P}_N \phi^k - \mathbf{P}_N \phi\ _U$	
penev_A_ EQ0043 (2.46)	008	0.995 S +1	$(\Phi_N)^{*+} = \sum_{r=1}^N \sigma_r^{-1} \mathbf{s}_r (\psi_r,)$	$(\Phi_N)^{*+} = \sum_{r=1}^N \sigma_r^{-1} \mathbf{s}_r (\psi_r,).$	
penev_A_ EQ0025 (2.28)	007	0.997 N +0	$\sigma \mathbf{s}_{\sigma} \equiv \Phi \psi_{\sigma}$	$\sigma \mathbf{s}_{\sigma} \equiv \Phi \psi_{\sigma}$	
penev_A_ EQ0037 (2.40)	008	0.997 N +0	$(\phi^t,\phi^t) = \sum_{\sigma>0} \sigma^2 a_{\sigma}^{t^2}.$	$(\phi^t,\phi^t) = \sum_{\sigma>0} \sigma^2 a_{\sigma}^{t^2}.$	
penev_A_ EQ0048 (2.51)	008	0.997 N +0	$\phi^{err} \equiv \phi - \phi^{rec} = (1 - \mathbf{P}_N) \phi$	$\phi^{err} \equiv \phi - \phi^{rec} = (1 - \mathbf{P}_N) \phi$	
penev_A_ EQ0033 (2.36)	007	0.997 S +1	$\ \mathbf{P}'\mathbf{s}\ _S^2 = \langle \mathbf{P}'\mathbf{s}, \mathbf{P}'\mathbf{s} \rangle = \sum_{\sigma>0} (\mathbf{s}_{\sigma}, \mathbf{s})^2.$	$\ \mathbf{P}'\mathbf{s}\ _S^2 = \langle \mathbf{P}'\mathbf{s}, \mathbf{P}'\mathbf{s} \rangle = \sum_{\sigma>0} (\mathbf{s}_{\sigma}, \mathbf{s})^2.$	
penev_A_ EQ0038 (2.41)	008	0.998 N -1	$(\psi_{\sigma}, \mathbf{R} \psi_{\sigma'}) = \langle \Phi \psi_{\sigma}, \Phi \psi_{\sigma'} \rangle = \sigma \sigma' \frac{1}{T} \sum_t a_{\sigma}^t a_{\sigma'}^t,$	$\langle \psi_{\sigma}, \mathbf{R} \psi_{\sigma'} \rangle = \langle \Phi \psi_{\sigma}, \Phi \psi_{\sigma'} \rangle = \sigma \sigma' \frac{1}{T} \sum_t a_{\sigma}^t a_{\sigma'}^t,$	
penev_A_ EQ0082 (4.24)	019	0.998 N +0	$\mathbf{K}^{-1} = \Phi^* \mathbf{O}^{*+} = \sum_{\mathbf{f}} \Psi_{\mathbf{f}} \sigma_{\mathbf{f}} (\Psi_{\mathbf{f}},).$	$\mathbf{K}^{-1} = \Phi^* \mathbf{O}^{*+} = \sum_{\mathbf{f}} \Psi_{\mathbf{f}} \sigma_{\mathbf{f}} (\Psi_{\mathbf{f}},).$	
penev_A_ EQ0029 (2.32)	007	0.998 N +0	$\Phi^{*+} = \sum_{\sigma>0} \sigma^{-1} \mathbf{s}_{\sigma} (\psi_{\sigma},).$	$\Phi^{*+} = \sum_{\sigma>0} \sigma^{-1} \mathbf{s}_{\sigma} (\psi_{\sigma},).$	
Conf. MathPix confidence: ≥0.9 0.5-0.9 <0.5 — no value Residual render vs scan (inkdrill): C component W weak S stable N noise K clean not measured					

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penev_A_ E00022 (2.25)	007	■ 0.998 ● K +0	$\mathrm{R}=\sum_{\sigma}\sigma^2\psi_{\sigma}\left(\psi_{\sigma},\right)$	$\mathbf{R}=\sum_{\sigma}\sigma^2\psi_{\sigma}\left(\psi_{\sigma},\right)$	$\mathbf{R}=\sum_{\sigma}\sigma^2\psi_{\sigma}(\psi_{\sigma},)$
penev_A_E00062 (4.4)	017	■ 0.999 ● N +0	$\sigma_{\mathbf{f}}^2\equiv\tilde{r}(\mathbf{f})\propto\frac{1}{ \mathbf{f} ^{\alpha}}$	$\sigma_{\mathbf{f}}^2\equiv\tilde{r}(\mathbf{f})\propto\frac{1}{ \mathbf{f} ^{\alpha}}$	$\sigma_{\mathbf{r}}^2\equiv\tilde{r}(\mathbf{f})\propto\frac{1}{ \mathbf{f} ^{\alpha}}$
penev_A_ E00045 (2.38)	008	■ 0.999 ● W +1	$\left(\Phi_N\right)^{*+}\phi=\sum_{r=1}^N\sigma_r^{-1}\mathbf{s}_r\left(\psi_r,\phi\right)=\sum_{r=1}^Na_r\mathbf{s}_r.$	$(\Phi_N)^{*+}\phi=\sum_{r=1}^N\sigma_r^{-1}\mathbf{s}_r\left(\psi_r,\phi\right)=\sum_{r=1}^Na_r\mathbf{s}_r.$	$(\Phi_N)^{*+}\phi=\sum_{r=1}^N\sigma_r^{-1}\mathbf{s}_r(\psi_r,\phi)=\sum_{r=1}^Na_r\mathbf{s}_r.$
penev_A_ E00012 (2.13)	006	■ 0.999 ● N +0	$\Phi\hat{\mathbf{x}}=\frac{1}{T}\sum_t\hat{t}\phi^t(\mathbf{x}).$	$\Phi\hat{\mathbf{x}}=\frac{1}{T}\sum_t\hat{t}\phi^t(\mathbf{x}).$	$\Phi\hat{\mathbf{x}}=\frac{1}{T}\sum_t\hat{t}\phi^t(\mathbf{x}).$
penev_A_ E00091 (4.32)	020	■ 0.999 ● N +0	$O(\mathbf{x})=(\hat{\mathbf{x}},O)=\sum_{r=1}^Na_r\Psi_r(\mathbf{x})$	$O(\mathbf{x})=(\hat{\mathbf{x}},O)=\sum_{r=1}^Na_r\Psi_r(\mathbf{x})$	$O(\mathbf{x})=(\hat{\mathbf{x}},O)=\sum_{r=1}^Na_r\Psi_r(\mathbf{x}).$
penev_A_E00007 (2.7)	006	■ 1.000 ● N +1	$s=\frac{1}{T}\sum_t\hat{t}s(t)\in S.$	$s=\frac{1}{T}\sum_t\hat{t}s(t)\in S.$	$\mathbf{s}=\frac{1}{T}\sum_t\hat{t}s(t)\in S.$
penev_A_ E00023 (2.26)	007	■ 1.000 ● N +0	$\mathbf{P}=\sum_{\sigma>0}\psi_{\sigma}\left(\psi_{\sigma},\right).$	$\mathbf{P}=\sum_{\sigma>0}\psi_{\sigma}\left(\psi_{\sigma},\right).$	$\mathbf{P}=\sum_{\sigma>0}\psi_{\sigma}(\psi_{\sigma},).$
penev_A_E00105	022	■ 1.000 ● S -1	$O^{\mathrm{rec}}(\mathbf{x})=O_m(\hat{\mathbf{x}},\mathbf{O}^*\mathbf{p}_{\mathbf{x}_m})=\frac{O_m}{\langle\mathbf{q}_{\mathbf{x}_m},\mathbf{q}_{\mathbf{x}_m}\rangle}(\hat{\mathbf{x}},\mathbf{O}^*\mathbf{q}_{\mathbf{x}_m})$	$O^{\mathrm{rec}}(\mathbf{x})=O_m(\hat{\mathbf{x}},\mathbf{O}^*\mathbf{p}_{\mathbf{x}_m})=\frac{O_m}{\langle\mathbf{q}_{\mathbf{x}_m},\mathbf{q}_{\mathbf{x}_m}\rangle}(\hat{\mathbf{x}},\mathbf{O}^*\mathbf{q}_{\mathbf{x}_m})$	$O^{\mathrm{rec}}(\mathbf{x})=O_m(\hat{\mathbf{x}},\mathbf{O}^*\mathbf{p}_{\mathbf{x}_m})=\frac{O_m}{\langle\mathbf{q}_{\mathbf{x}_m},\mathbf{q}_{\mathbf{x}_m}\rangle}(\hat{\mathbf{x}},\mathbf{O}^*\mathbf{q}_{\mathbf{x}_m}).$
penev_A_ E00110 (4.51)	022	■ 1.000 ● C +3	$\phi^{\mathrm{rec}}(\mathbf{x})=O_m\frac{K^{(-1)}\mathbf{x}_m(\mathbf{x})}{P_{\mathbf{x}_m}(\mathbf{x}_m)}$	$\phi^{\mathrm{rec}}(\mathbf{x})=O_m\frac{K^{(-1)}\mathbf{x}_m(\mathbf{x})}{P_{\mathbf{x}_m}(\mathbf{x}_m)}$	$\phi^{\mathrm{rec}}(\mathbf{x})=O_m\frac{K^{(-1)}\mathbf{x}_m(\mathbf{x})}{P_{\mathbf{x}_m}(\mathbf{x}_m)}$
penev_A_E00002 (2.2)	006	■ 1.000 ● N +2	$\mathcal{P}[\phi(\mathbf{x})]\propto\exp\left\{-\frac{1}{2}\int_Vd\mathbf{x}\phi(\mathbf{x})\int_Vd\mathbf{y}R^{-1}(\mathbf{x},\mathbf{y})\phi(\mathbf{y})\right\}$	$\mathcal{P}[\phi(\mathbf{x})]\propto\exp\left\{-\frac{1}{2}\int_Vd\mathbf{x}\phi(\mathbf{x})\int_Vd\mathbf{y}R^{-1}(\mathbf{x},\mathbf{y})\phi(\mathbf{y})\right\}$	$\mathcal{P}[\phi(\mathbf{x})]\propto\exp\{-\frac{1}{2}\int_Vd\mathbf{x}\phi(\mathbf{x})\int_Vd\mathbf{y}R^{-1}(\mathbf{x},\mathbf{y})\phi(\mathbf{y})\}.$
penev_A_E00003 (2.3)	006	■ 1.000 ● N +1	$\Phi:U\mapsto S$	$\Phi:U\mapsto S$	$\Phi:U\mapsto S$
penev_A_E00006 (2.6)	006	■ 1.000 ● N +1	$\phi=\frac{1}{V}\sum_{\mathbf{x}}\hat{\mathbf{x}}\phi(\mathbf{x})\in U$	$\phi=\frac{1}{V}\sum_{\mathbf{x}}\hat{\mathbf{x}}\phi(\mathbf{x})\in U$	$\phi=\frac{1}{V}\sum_{\mathbf{x}}\hat{\mathbf{x}}\phi(\mathbf{x})\in U$
penev_A_ E00010 (2.11)	006	■ 1.000 ● N +1	$\langle\mathbf{s},\Phi\mathbf{u}\rangle=(\Phi^*\mathbf{s},\mathbf{u}).$	$\langle\mathbf{s},\Phi\mathbf{u}\rangle=(\Phi^*\mathbf{s},\mathbf{u}).$	$\langle\mathbf{s},\Phi\mathbf{u}\rangle=(\Phi^*\mathbf{s},\mathbf{u}).$
penev_A_ E00013 (2.14)	006	■ 1.000 ● N +2	$\langle\Phi\hat{\mathbf{x}},\Phi\hat{\mathbf{y}}\rangle=\frac{1}{T}\sum_t\phi^t(\mathbf{x})\phi^t(\mathbf{y})$	$\langle\Phi\hat{\mathbf{x}},\Phi\hat{\mathbf{y}}\rangle=\frac{1}{T}\sum_t\phi^t(\mathbf{x})\phi^t(\mathbf{y})$	$\langle\Phi\hat{\mathbf{x}},\Phi\hat{\mathbf{y}}\rangle=\frac{1}{T}\sum_t\phi^t(\mathbf{x})\phi^t(\mathbf{y})$
penev_A_ E00014 (2.15)	006	■ 1.000 ● N +1	$\Phi^*\hat{t}\equiv\mathbf{1}_U\Phi^*\hat{t}=\frac{1}{V}\sum_{\mathbf{x}}\hat{\mathbf{x}}\phi^t(\mathbf{x})=\phi^t.$	$\Phi^*\hat{t}\equiv\mathbf{1}_U\Phi^*\hat{t}=\frac{1}{V}\sum_{\mathbf{x}}\hat{\mathbf{x}}\phi^t(\mathbf{x})=\phi^t.$	$\Phi^*\hat{t}\equiv\mathbf{1}_U\Phi^*\hat{t}=\frac{1}{V}\sum_{\mathbf{x}}\hat{\mathbf{x}}\phi^t(\mathbf{x})=\phi^t.$
penev_A_ E00015 (2.16)	007	■ 1.000 ● S -1	$\mathcal{P}\left[\Phi^*\mathbf{s}\right]\propto\exp\left\{-\frac{1}{2}\ \mathbf{s}\ _S^2\right\}$	$\mathcal{P}\left[\Phi^*\mathbf{s}\right]\propto\exp\left\{-\frac{1}{2}\ \mathbf{s}\ _S^2\right\}$	$\mathcal{P}[\Phi^*\mathbf{s}]\propto\exp\{-\frac{1}{2}\ \mathbf{s}\ _S^2\}$
penev_A_ E00016 (2.17)	007	■ 1.000 ● K +0	$\Phi^{*+}:U\mapsto S$	$\Phi^{*+}:U\mapsto S$	$\Phi^{*+}:U\mapsto S$
Conf. MathPix confidence: ■ ≥ 0.9 ■ 0.5–0.9 ■ < 0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured					

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penev_A_ EQ0017 (2.18)	007	■ 1.000 ● N +0	$\mathrm{\Phi^*\Phi^*+\Phi^*\Phi^*\mathrm{\Phi^*}}=\mathrm{\Phi^*\Phi^*}$	$\Phi^*\Phi^{*+}\Phi^*\mathbf{s}=\Phi^*\mathbf{s}.$	$\Phi^*\Phi^{*+}\Phi^*\mathbf{s}=\Phi^*\mathbf{s}.$
penev_A_ EQ0019 (2.22)	007	■ 1.000 ● N +1	$\left(\mathrm{\mathbf{u}},\mathrm{\Phi^*\Phi^*\mathbf{u}}\right)=\left(\mathrm{\Phi^*\mathbf{u}},\mathrm{\Phi^*\mathbf{u}}\right)=\left\ \mathrm{\Phi^*\mathbf{u}}\right\ _S^2$	$(\mathbf{u},\Phi^*\Phi\mathbf{u})=\langle\Phi\mathbf{u},\Phi\mathbf{u}\rangle=\ \Phi\mathbf{u}\ _S^2$	$(\mathbf{u},\Phi^*\Phi\mathbf{u})=\langle\Phi\mathbf{u},\Phi\mathbf{u}\rangle=\ \Phi\mathbf{u}\ _S^2$
penev_A_ EQ0036 (2.39)	008	■ 1.000 ● N +1	$\mathrm{\phi^t}=\sum_{\sigma>0}\sigma\mathrm{a}_{\sigma}^t\mathrm{\psi}_{\sigma}.$	$\phi^t=\sum_{\sigma>0}\sigma a_{\sigma}^t\psi_{\sigma}.$	$\phi^t=\sum_{\sigma>0}\sigma a_{\sigma}^t\psi_{\sigma}.$
penev_A_ EQ0046 (2.49)	008	■ 1.000 ● W -2	$\mathrm{\mathcal{P}[\phi]}\propto\mathrm{exp}\left\{-\frac{1}{2}\sum_{r=1}^N\mathrm{a}_r^2\right\}\propto\prod_{r=1}^N\mathrm{exp}\left\{-\frac{1}{2}\mathrm{a}_r^2\right\}$	$\mathcal{P}[\phi]\propto\exp\left\{-\frac{1}{2}\sum_{r=1}^Na_r^2\right\}\propto\prod_{r=1}^N\exp\left\{-\frac{1}{2}a_r^2\right\}$	$\mathcal{P}[\phi]\propto\exp\left\{-\frac{1}{2}\sum_{r=1}^Na_r^2\right\}\propto\prod_{r=1}^N\exp\left\{-\frac{1}{2}a_r^2\right\}$
penev_A_EQ0053 (3.4)	014	■ 1.000 ● N +0	$\mathrm{\mu}\left(\mathrm{d}^{(p,q)}\right)=\sum_{r=1}^Nw_r\left(\mathrm{a}_r^p-\mathrm{a}_r^q\right)^2=\sum_{r=1}^Nw_rD_r^{(p,q)}\equiv D^{(p,q)}$	$\mu\left(\mathbf{d}^{(p,q)}\right)=\sum_{r=1}^Nw_r\left(a_r^p-a_r^q\right)^2=\sum_{r=1}^Nw_rD_r^{(p,q)}\equiv D^{(p,q)}$	$\mu(\mathbf{d}^{(\mathfrak{p},q)})=\sum_{r=1}^Nw_r(a_r^p-a_r^q)^2=\sum_{r=1}^Nw_rD_r^{(p,q)}\equiv D^{(p,q)}$
penev_A_EQ0054 (3.5)	014	■ 1.000 ● W -1	$\mathrm{D}_{\mathrm{r}}^{\left(\mathrm{p},\mathrm{q}\right)}\equiv\left(\mathrm{a}_{\mathrm{r}}^{\mathrm{p}}-\mathrm{a}_{\mathrm{r}}^{\mathrm{q}}\right)^2.$	$D_r^{(p,q)}\equiv(a_r^p-a_r^q)^2.$	$D_r^{(\mathfrak{p},q)}\equiv(a_r^p-a_r^q)^2.$
penev_A_EQ0058 (3.9)	015	■ 1.000 ● W -1	$w_r^{\mathrm{naive}}\propto\left(m_r^{\mathrm{same}}-m_r^{\mathrm{diff}}\right)\left(H^{\mathrm{intra}}\right)_{rr}^{-1}.$	$w_r^{\mathrm{naive}}\propto\left(m_r^{\mathrm{same}}-m_r^{\mathrm{diff}}\right)\left(H^{\mathrm{intra}}\right)_{rr}^{-1}.$	$w_r^{\mathrm{naive}}\propto\left(m_r^{\mathrm{same}}-m_r^{\mathrm{diff}}\right)\left(H^{\mathrm{intra}}\right)_{rr}^{-1}.$
penev_A_EQ0059 (4.1)	017	■ 1.000 ● N +1	$R(\mathrm{\mathbf{x}},\mathrm{\mathbf{y}})=r(\mathrm{\mathbf{x}}-\mathrm{\mathbf{y}}).$	$R(\mathbf{x},\mathbf{y})=r(\mathbf{x}-\mathbf{y}).$	$R(\mathbf{x},\mathbf{y})=r(\mathbf{x}-\mathbf{y}).$
penev_A_EQ0060 (4.2)	017	■ 1.000 ● N +0	$R(\mathrm{\mathbf{x}},\mathrm{\mathbf{y}})=r(\mathrm{\mathbf{x}}-\mathrm{\mathbf{y}})=\rho(\ell).$	$R(\mathbf{x},\mathbf{y})=r(\mathbf{x}-\mathbf{y})=\rho(\ell).$	$R(\mathbf{x},\mathbf{y})=\tau(\mathbf{x}-\mathbf{y})=\rho(\ell).$
penev_A_EQ0064 (4.6)	018	■ 1.000 ● N +2	$\mathrm{\mathbf{K}}\colon U\rightarrow U$	$\mathbf{K}:U\rightarrow U$	$\mathbf{K}:U\rightarrow U$
penev_A_ EQ0068 (4.10)	018	■ 1.000 ● S +1	$\left\langle O(\mathbf{x})O(\mathbf{y})\right\rangle_T=\frac{1}{T}\sum_tO^t(\mathbf{x})O^t(\mathbf{y})\propto\delta_{\mathbf{x},\mathbf{y}}$	$\langle O(\mathbf{x})O(\mathbf{y})\rangle_T=\frac{1}{T}\sum_tO^t(\mathbf{x})O^t(\mathbf{y})\propto\delta_{\mathbf{x},\mathbf{y}}$	$\langle O(\mathbf{x})O(\mathbf{y})\rangle_T=\frac{1}{T}\sum_tO^t(\mathbf{x})O^t(\mathbf{y})\propto\delta_{\mathbf{x},\mathbf{y}}$
penev_A_ EQ0089 (4.30)	020	■ 1.000 ● C +3	$\mathrm{\mathbf{K}}^{\left(n\right)}=\sum_{r=1}^N\psi_r\left(\frac{1}{\sigma_r}\right)^n\left(\psi_r,\right).$	$\mathbf{K}^{(n)}=\sum_{r=1}^N\psi_r\left(\frac{1}{\sigma_r}\right)^n\left(\psi_r,\right).$	$\mathbf{K}^{(n)}=\sum_{r=1}^N\psi_r\left(\frac{1}{\sigma_r}\right)^n\left(\psi_r,\right).$
penev_A_ EQ0102 (4.43)	022	■ 1.000 ● S -1	$\left(\mathrm{\mathbf{q}}_{\mathbf{x}_m},\mathrm{\mathbf{s}}_{\perp}\right)=\left(\mathrm{\mathbf{q}}_{\mathbf{x}_m},\mathrm{\mathbf{s}}-\mathrm{\mathbf{s}}_{\parallel}\right)=\left(\mathrm{\mathbf{q}}_{\mathbf{x}_m},\mathrm{\mathbf{s}}\right)-\left(\mathrm{\mathbf{q}}_{\mathbf{x}_m},\mathrm{\mathbf{s}}_{\parallel}\right)=O_m-O_m=0.$	$\langle\mathbf{q}_{\mathbf{x}_m},\mathbf{s}_{\perp}\rangle=\langle\mathbf{q}_{\mathbf{x}_m},\mathbf{s}-\mathbf{s}_{\parallel}\rangle=\langle\mathbf{q}_{\mathbf{x}_m},\mathbf{s}\rangle-\langle\mathbf{q}_{\mathbf{x}_m},\mathbf{s}_{\parallel}\rangle=O_m-O_m=0.$	$\langle\mathbf{q}_{\mathbf{x}_m},\mathbf{s}_{\perp}\rangle=\langle\mathbf{q}_{\mathbf{x}_m},\mathbf{s}-\mathbf{s}_{\parallel}\rangle=\langle\mathbf{q}_{\mathbf{x}_m},\mathbf{s}\rangle-\langle\mathbf{q}_{\mathbf{x}_m},\mathbf{s}_{\parallel}\rangle=O_m-O_m=0.$
penev_A_ EQ0109 (4.50)	022	■ 1.000 ● W -2	$\mathrm{\phi^{\mathrm{rec}}(\mathbf{x})}=\left(\hat{\mathbf{x}},\mathrm{\Phi^*\mathbf{s}^{\mathrm{rec}}}\right)$	$\phi^{\mathrm{rec}}(\mathbf{x})=(\hat{\mathbf{x}},\Phi^*\mathbf{s}^{\mathrm{rec}})$	$\phi^{\mathrm{rec}}(\mathbf{x})=(\hat{\mathbf{x}},\Phi^*\mathbf{s}^{\mathrm{rec}})$
penev_A_ EQ0116 (4.57)	022	■ 1.000 ● C +6	$\left(\mathrm{\mathbf{q}}_m,\mathrm{\mathbf{s}}_{\perp}\right)=\left(\mathrm{\mathbf{q}}_m,\mathrm{\mathbf{s}}-\mathrm{\mathbf{s}}_{\parallel}\right)=\left(\mathrm{\mathbf{q}}_m,\mathrm{\mathbf{s}}\right)-\left(\mathrm{\mathbf{q}}_m,\mathrm{\mathbf{s}}_{\parallel}\right)=O_m-O_m=0.$	$\langle\mathbf{q}_m,\mathbf{s}_{\perp}\rangle=\langle\mathbf{q}_m,\mathbf{s}-\mathbf{s}_{\parallel}\rangle=\langle\mathbf{q}_m,\mathbf{s}\rangle-\langle\mathbf{q}_m,\mathbf{s}_{\parallel}\rangle=O_m-O_m=0.$	$\langle\mathbf{q}_m,\mathbf{s}_{\perp}\rangle=\langle\mathbf{q}_m,\mathbf{s}-\mathbf{s}_{\parallel}\rangle=\langle\mathbf{q}_m,\mathbf{s}\rangle-\langle\mathbf{q}_m,\mathbf{s}_{\parallel}\rangle=O_m-O_m=0.$
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured					

Tables

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image																																																																																																																				
penev_A_TAB_001	014	—	<pre>\begin{tabular}[t]{ l l l l }\hline\multicolumn{4}{ c }{43}\hline\hline\frac{n(n-1)}{2}\) & \((s(n)\) & \(\frac{n(n-1)}{2}\)\) & \((s(n)\times\frac{n(n-1)}{2}\)\)\hline1 & 26 & 0 & 0\hline2 & 769 & 1 & 769\hline3 & 145 & 3 & 435\hline4 & 143 & 6 & 858\hline5 & 12 & 10 & 120\hline6 & 17 & 15 & 255\hline7 & 8 & 21 & 168\hline8 & 3 & 28 & 84\hline9 & 2 & 36 & 72\hline10 & 3 & 45 & 135\hline11 & 16 & 55 & 880\hline12 & 1 & 67 & 67\hline& & & \((N_s=3842)\)\hline\end{tabular}</pre>	<table><tr><th colspan="4">43</th></tr><tr><th>n</th><th>$s(n)$</th><th>$\frac{n(n-1)}{2}$</th><th>$s(n)\times\frac{n(n-1)}{2}$</th></tr><tr><td>1</td><td>26</td><td>0</td><td>0</td></tr><tr><td>2</td><td>769</td><td>1</td><td>769</td></tr><tr><td>3</td><td>145</td><td>3</td><td>435</td></tr><tr><td>4</td><td>143</td><td>6</td><td>858</td></tr><tr><td>5</td><td>12</td><td>10</td><td>120</td></tr><tr><td>6</td><td>17</td><td>15</td><td>255</td></tr><tr><td>7</td><td>8</td><td>21</td><td>168</td></tr><tr><td>8</td><td>3</td><td>28</td><td>84</td></tr><tr><td>9</td><td>2</td><td>36</td><td>72</td></tr><tr><td>10</td><td>3</td><td>45</td><td>135</td></tr><tr><td>11</td><td>16</td><td>55</td><td>880</td></tr><tr><td>12</td><td>1</td><td>67</td><td>67</td></tr><tr><td></td><td></td><td></td><td>$N_s = 3842$</td></tr></table>	43				n	$s(n)$	$\frac{n(n-1)}{2}$	$s(n)\times\frac{n(n-1)}{2}$	1	26	0	0	2	769	1	769	3	145	3	435	4	143	6	858	5	12	10	120	6	17	15	255	7	8	21	168	8	3	28	84	9	2	36	72	10	3	45	135	11	16	55	880	12	1	67	67				$N_s = 3842$	<table><tr><th>n</th><th>$s(n)$</th><th>$\frac{n(n-1)}{2}$</th><th>$s(n)\times\frac{n(n-1)}{2}$</th></tr><tr><td>1</td><td>26</td><td>0</td><td>0</td></tr><tr><td>2</td><td>769</td><td>1</td><td>769</td></tr><tr><td>3</td><td>145</td><td>3</td><td>435</td></tr><tr><td>4</td><td>143</td><td>6</td><td>858</td></tr><tr><td>5</td><td>12</td><td>10</td><td>120</td></tr><tr><td>6</td><td>17</td><td>15</td><td>255</td></tr><tr><td>7</td><td>8</td><td>21</td><td>168</td></tr><tr><td>8</td><td>3</td><td>28</td><td>84</td></tr><tr><td>9</td><td>2</td><td>36</td><td>72</td></tr><tr><td>10</td><td>3</td><td>45</td><td>135</td></tr><tr><td>11</td><td>16</td><td>55</td><td>880</td></tr><tr><td>12</td><td>1</td><td>67</td><td>67</td></tr><tr><td></td><td></td><td></td><td>$N_s = 3842$</td></tr></table>	n	$s(n)$	$\frac{n(n-1)}{2}$	$s(n)\times\frac{n(n-1)}{2}$	1	26	0	0	2	769	1	769	3	145	3	435	4	143	6	858	5	12	10	120	6	17	15	255	7	8	21	168	8	3	28	84	9	2	36	72	10	3	45	135	11	16	55	880	12	1	67	67				$N_s = 3842$
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Conf. MathPix confidence:

■

 ≥0.9

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 0.5-0.9

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 <0.5

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 no value

Residual render vs scan (inkdrill):

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 C component

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 W weak

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 S stable

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 N noise

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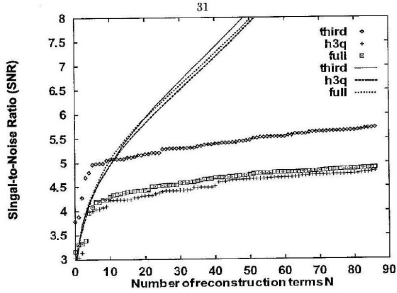
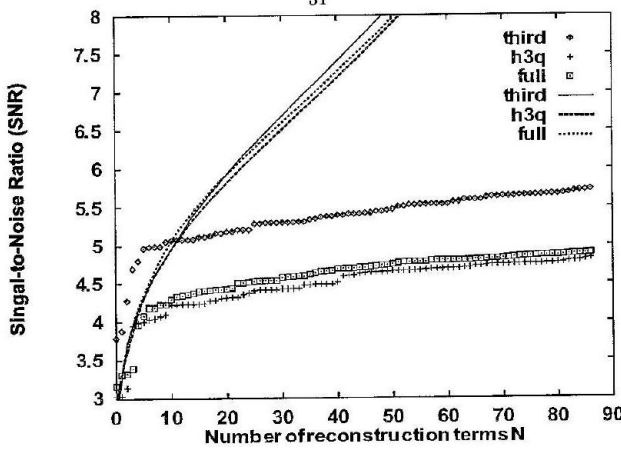
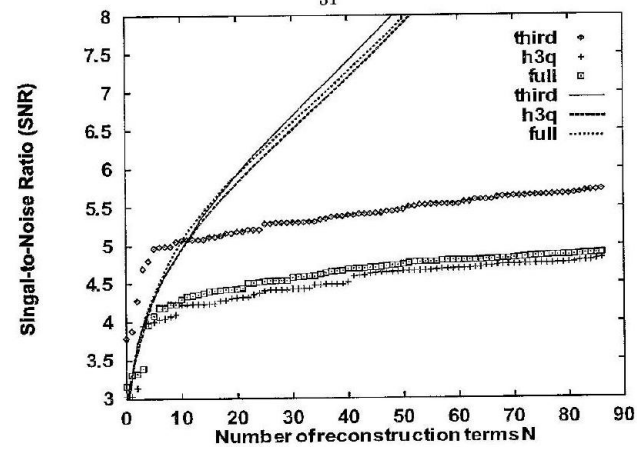
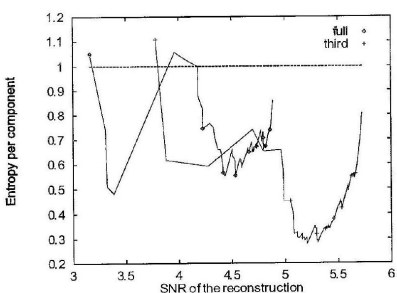
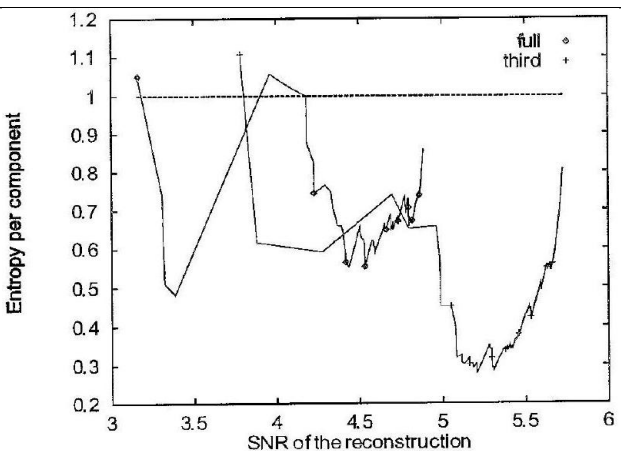
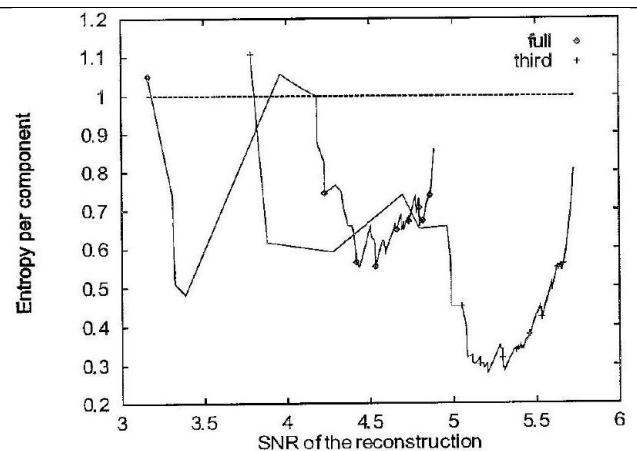
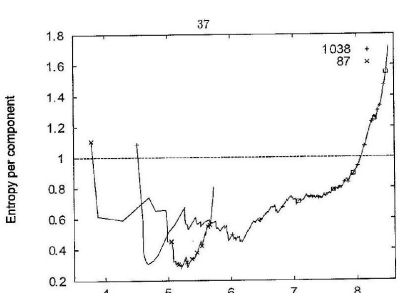
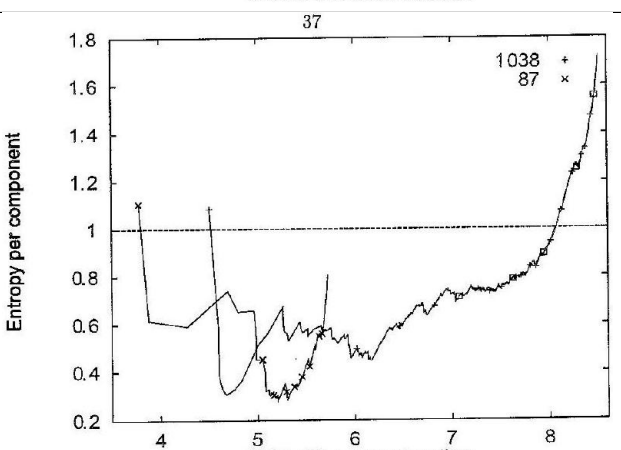
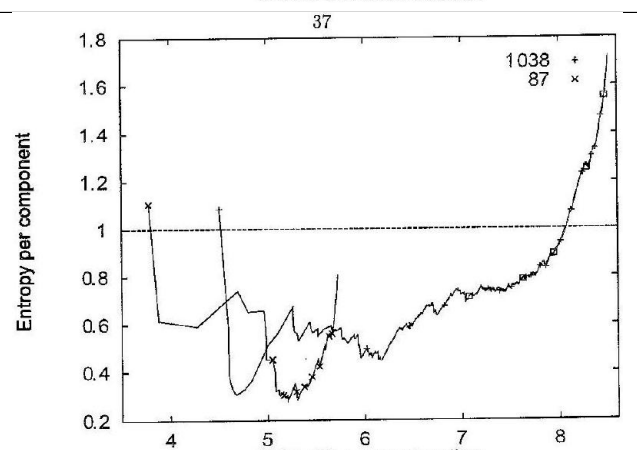
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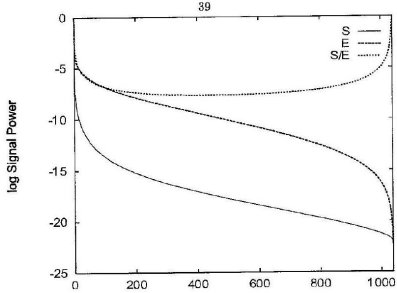
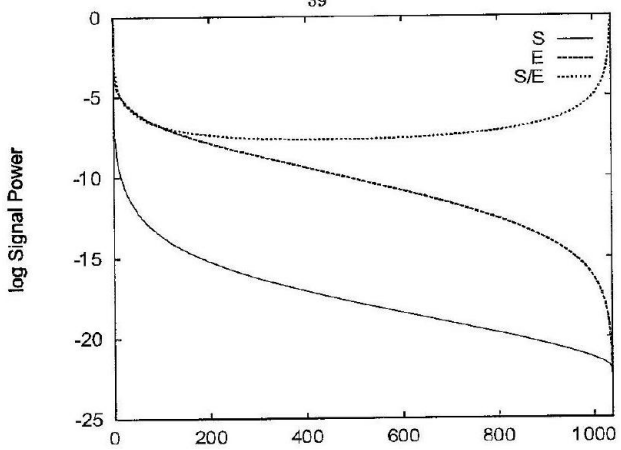
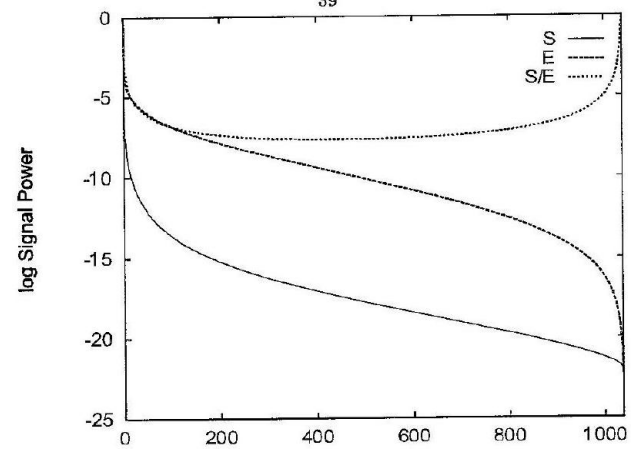
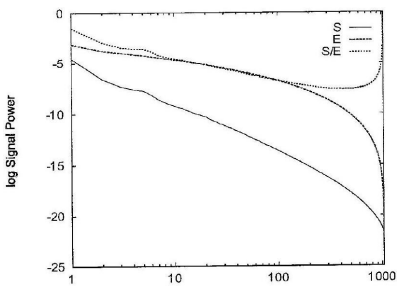
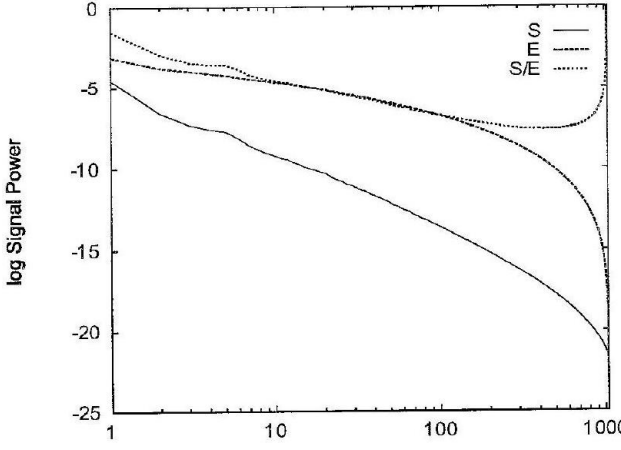
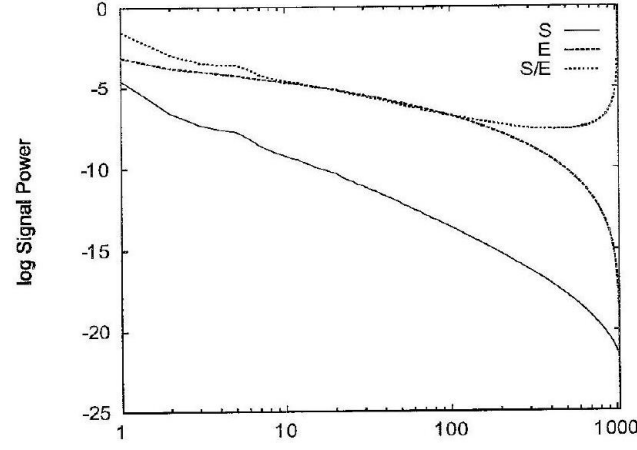
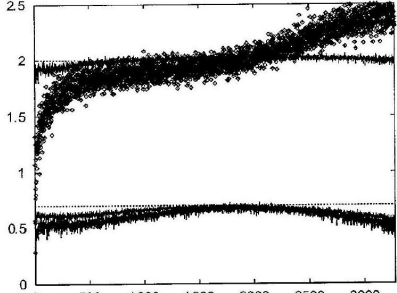
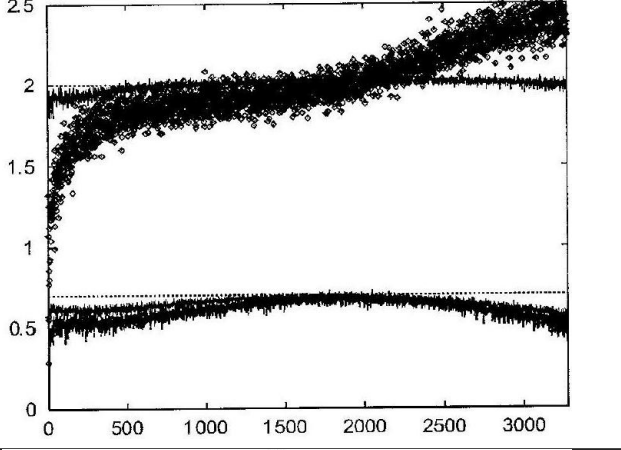
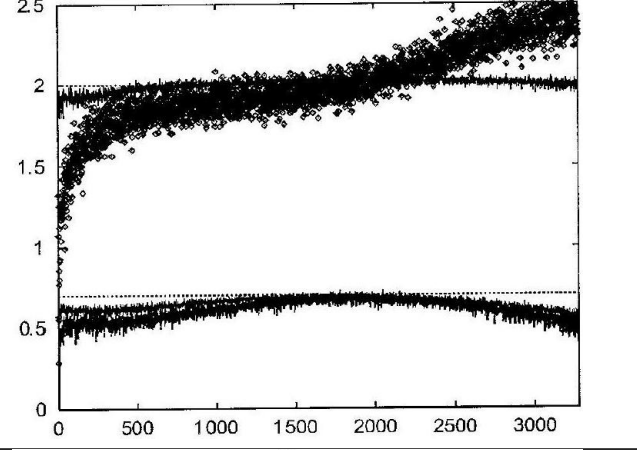
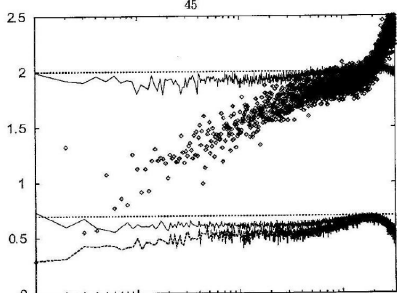
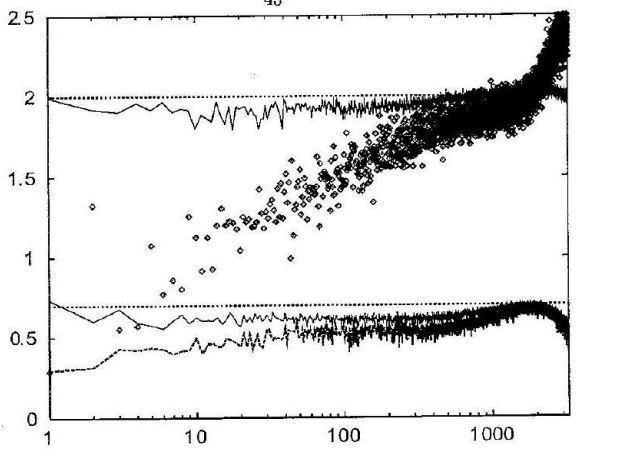
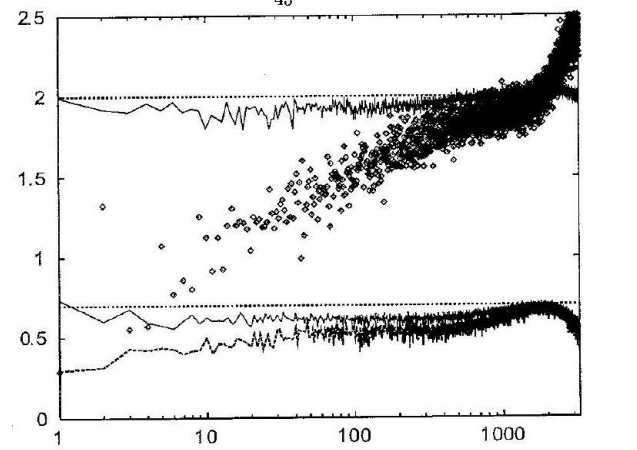
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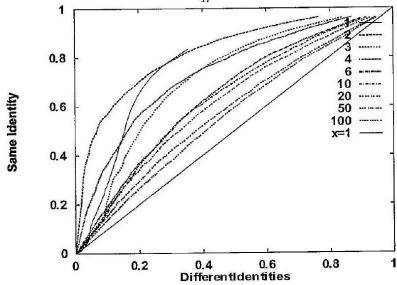
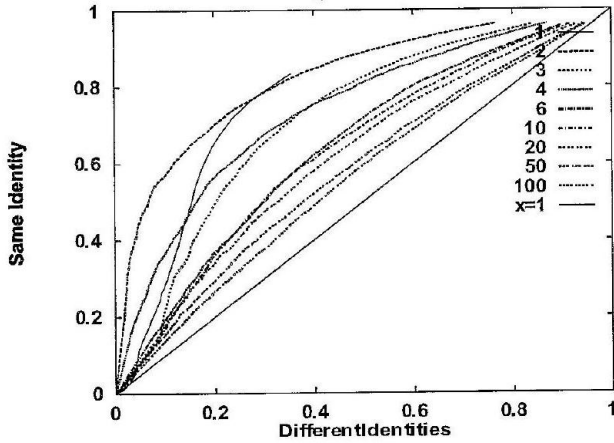
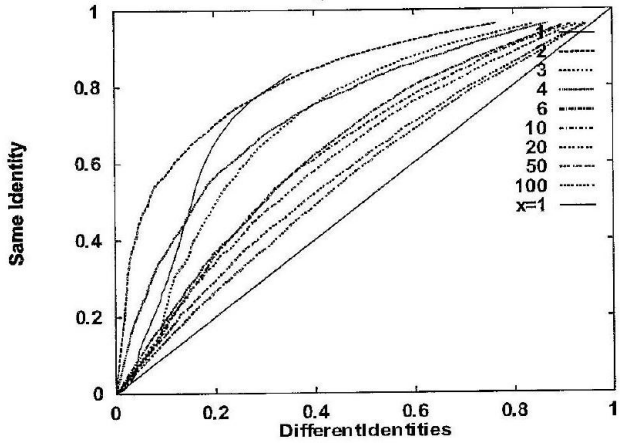
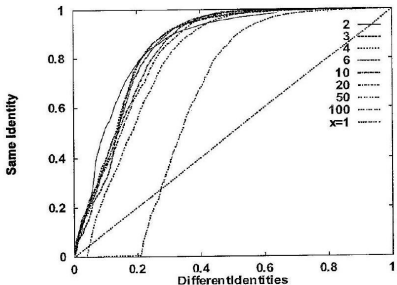
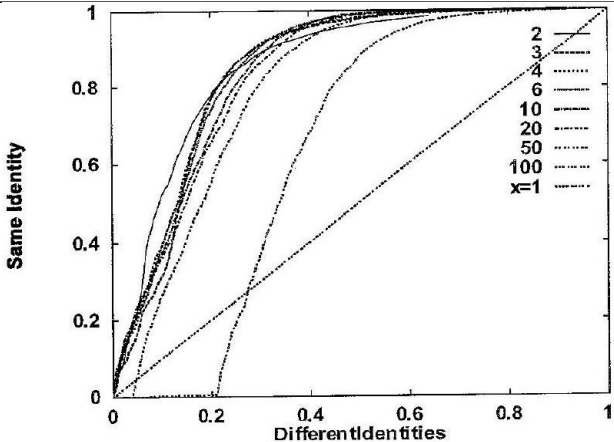
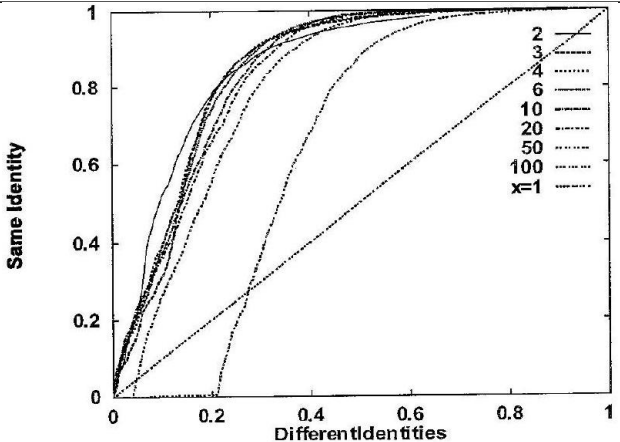
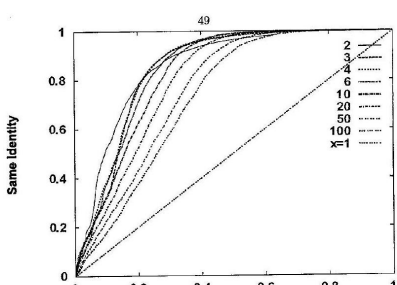
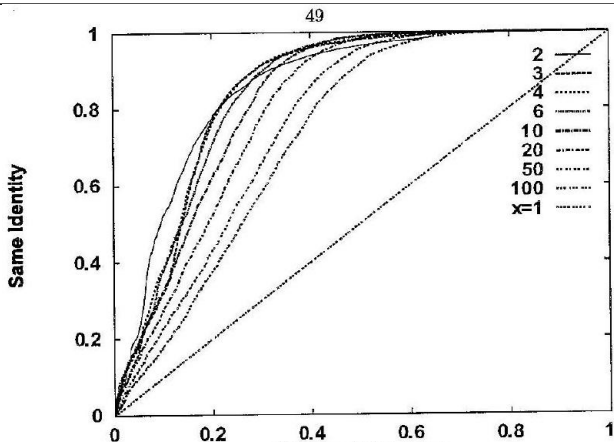
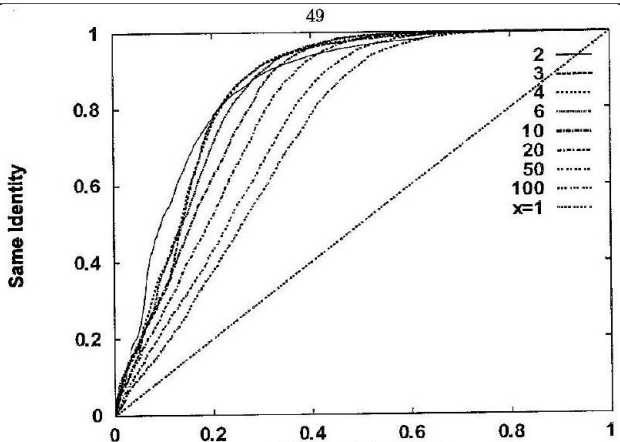
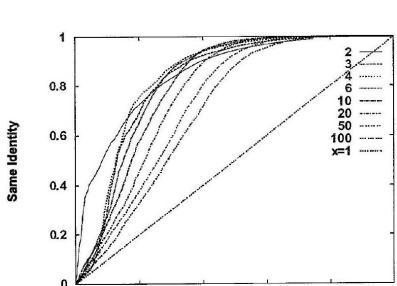
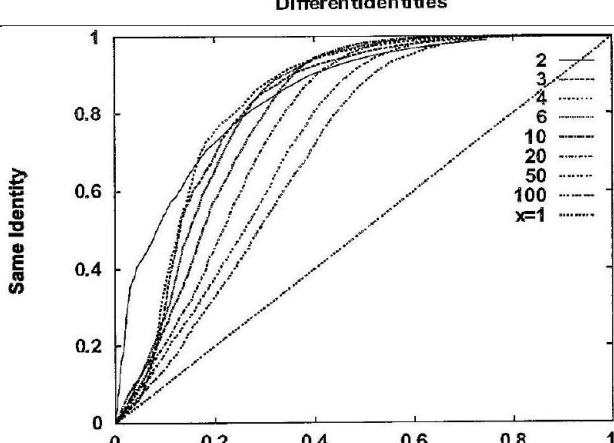
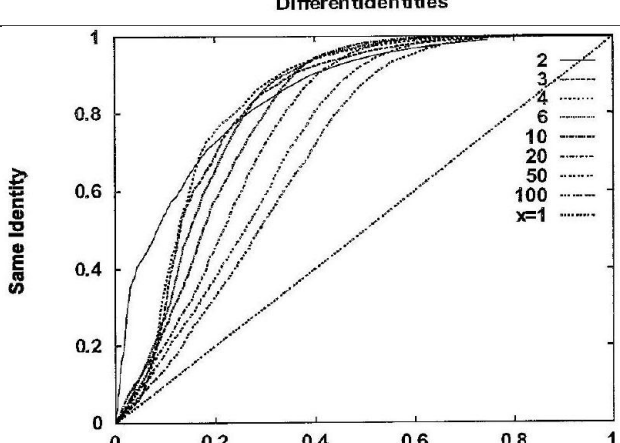
 not measured

Image regions — rendered against scan

Two comparable cells per row. **Rendered**: the region’s own LaTeX compiled as its OWN document, or — where no LaTeX exists — the scan again, so every row has two cells and the ink difference reads as a floor rather than as missing-versus-present. A duplicated row is marked (*dup*) and its distance is a SELF-comparison, not agreement between two sources. **Scan**: the tex.zip image whose filename region 5-tuple matches this row, else the CDN crop. The Class column carries the residual class beside MathPix’s confidence, as the equation rows do. Setting a region’s LaTeX does NOT say it renders to what is on the page — that is what the two columns are for, and 281 is the open question they exist to let a reader answer. A region with no LaTeX can be reconstructed with pdfdrill vision; verify any LLM result against the real ink with inkdrill.

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penev_A_PIC_0003	012	— (dup)	4543005f-bb5d-4491-a9d4-2522beda7f5-12_594_795_1435_1078.jpg			

penev_A_PIC_0004	013	— (dup)	4543005f-bb5d-4491-a9d4-2522beda47f5-13_599_804_1441_139.jpg			
penev_A_PIC_0005	013	— (dup)	4543005f-bb5d-4491-a9d4-2522beda47f5-13_578_818_94_1065.jpg			
penev_A_PIC_0006	014	— (dup)	4543005f-bb5d-4491-a9d4-2522beda47f5-14_580_742_91_1117.jpg			
penev_A_PIC_0007	014	— (dup)	4543005f-bb5d-4491-a9d4-2522beda47f5-14_599_742_1432_1126.jpg			

penev_A_PIC_0008	015	— (dup)	4543005f-bb5d-4491-a9d4-2522beda47f5-15_599_802_1439_139.jpg			
penev_A_PIC_0009	015	— (dup)	4543005f-bb5d-4491-a9d4-2522beda47f5-15_573_800_94_1065.jpg			
penev_A_PIC_0010	015	— (dup)	4543005f-bb5d-4491-a9d4-2522beda47f5-15_599_800_1430_1074.jpg			
penev_A_PIC_0011	016	— (dup)	4543005f-bb5d-4491-a9d4-2522beda47f5-16_580_802_98_132.jpg			

penev_A_DIA_0001	009	— (dup)	4543005f-bb5d-4491-a9d4-2522beda47f5-09_708_870_196_1037.jpg		
penev_A_DIA_0002	009	— (dup)	4543005f-bb5d-4491-a9d4-2522beda47f5-09_692_873_1435_1046.jpg		
penev_A_DIA_0003	010	— (dup)	4543005f-bb5d-4491-a9d4-2522beda47f5-10_361_873_1439_112.jpg		
penev_A_DIA_0004	010	— (dup)	4543005f-bb5d-4491-a9d4-2522beda47f5-10_247_875_85_1033.jpg		

penev_A_DIA_0005	010	— (dup)	4543005f-bb5d-4491-a9d4-2522beda47f5-10_610_870_1430_1049.jpg			
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penev_A_DIA_0007	012	— (dup)	4543005f-bb5d-4491-a9d4-2522beda47f5-12_439_873_98_103.jpg			
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penev_A_DIA_0014	021	— (dup)	4543005f-bb5d-4491-a9d4-2522beda47f5-21_281_870_91_107.jpg				